



T.C.

HALIC UNIVERSITY

GRADUATE EDUCATION INSTITUTE

**Natural Terpenes vs Synthetic Pesticides: A Comparative Docking Study on
Acetylcholinesterase**

Comparison of Limonene and L-Carvone with Carbaryl and Malathion

Miray CEYLAN

DEPARTMENT OF MOLECULAR BIOLOGY AND GENETICS

UNDERGRADUATE THESIS

SUPERVISOR

Prof. Dr. Lecturer Mehmet Emin URAS

ISTANBUL – 2026

ETHICAL DECLARATION

I hereby declare that this undergraduate thesis entitled “Natural Terpenes vs Synthetic Pesticides: A Comparative Docking Study on Acetylcholinesterase: Comparison of Limonene and L-Carvone with Carbaryl and Malathion” has been prepared by me under the supervision of Prof. Dr. Lecturer Mehmet Emin Uras at Haliç University, Department of Molecular Biology and Genetics.

I declare that all stages of this study were carried out in accordance with scientific research principles and ethical rules. I further declare that all information, data, figures, tables, and sources used in this thesis have been properly cited and listed in the references section, and that no part of this work has been presented as my own without appropriate acknowledgment.

I accept full responsibility for any situation that may arise in the event of a contrary finding.

Miray CEYLAN
June, 2026

ABSTRACT

Natural Terpenes vs Synthetic Pesticides: A Comparative Docking Study on
Acetylcholinesterase

Comparison of Limonene and L-Carvone with Carbaryl and Malathion

Halic University

Institute of Graduate Studies

Department of Molecular biology and Genetic, Undergraduate Thesis

Supervisor: Assist Prof. Dr. Mehmet Emin Uras

June, 2026, 49 pages

This thesis investigated the predicted interaction profiles of two plant-derived monoterpenes, limonene and L-carvone, and two synthetic pesticide compounds, carbaryl and malathion, against acetylcholinesterase (AChE). A computational docking-based approach was used to compare these compounds under the same target conditions. AChE was chosen because of its essential function in cholinergic signal regulation and its well-known relevance in pesticide-related neurotoxicity. The three-dimensional enzyme structure was obtained from the Protein Data Bank using PDB ID 6XYY, and the ligand structures were retrieved from PubChem. Docking calculations were performed with AutoDock Vina, while the outputs were interpreted using binding affinity, RMSD values, estimated K_i values, ADMET-related predictions, and residue-level protein–ligand interaction analysis.

The docking results showed that carbaryl had the strongest predicted interaction with AChE, with a binding energy of -8.489 kcal/mol and an estimated K_i value of 0.60 μ M. Among the natural compounds, L-carvone produced a more favorable docking result than limonene, with a binding energy of -6.675 kcal/mol and an estimated K_i value of 12.88 μ M. Malathion showed a binding energy close to L-carvone; however, its predicted ADMET-related behavior was different because of its higher TPSA value, greater molecular flexibility, low predicted gastrointestinal absorption, and absence of predicted blood–brain barrier permeability.

Interaction analysis further supported the docking interpretation. Limonene, L-carvone, and carbaryl shared Tyr71, Trp83, Tyr370, and Trp472 with the native ligand, whereas malathion shared Tyr71, Tyr370, Phe371, and His480. The combined use of PyMOL, PLIP, and Discovery Studio Visualizer allowed the numerical docking results to be interpreted together with active-site residue interactions. Overall, carbaryl showed the strongest predicted AChE binding profile, while L-carvone appeared to be the most promising compound among the natural monoterpenes evaluated in this study. Since the findings are based on computational predictions, experimental enzyme inhibition and toxicity studies are required before drawing final conclusions about the practical pesticide-related potential of limonene and L-carvone.

Keywords: Acetylcholinesterase, molecular docking, limonene, L-carvone, carbaryl, malathion, AutoDock Vina, SwissADME

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my thesis supervisor, Prof. Dr. Lecturer Mehmet Emin Uras, for his valuable guidance, support, patience, and constructive feedback throughout the preparation of this undergraduate thesis. His academic knowledge and suggestions greatly helped me during the development and completion of this study.

I would also like to thank Haliç University, Department of Molecular Biology and Genetics, for providing me with the academic background and learning environment that supported my undergraduate education.

I am deeply grateful to my family for their continuous encouragement, patience and support throughout my education. Their moral support and belief in me have always been a great source of motivation.

Finally, I would like to thank everyone who supported me directly or indirectly during this thesis process.

Miray CEYLAN
June, 2026

TABLE OF CONTENTS

	<u>Page</u>
ABSTRACT.....	iii
ACKNOWLEDGEMENTS.....	iii
TABLE OF CONTENTS.....	iii
LIST OF TABLES.....	v
LIST OF FIGURES.....	vi
LIST OF ABBREVIATIONS.....	viii
LIST OF SYMBOLS.....	ix
 1. INTRODUCTION.....	 1
1.1 Pesticides and Their Biological Effects.....	1
1.2 Neurotoxicity of Pesticides.....	3
1.3 Acetylcholinesterase (AChE) Enzyme and Its Role.....	5
1.4 Pesticide-Induced Diseases.....	8
1.5 Natural Compounds as Alternative Pesticides.....	10
1.6 Molecular Docking in Drug and Pesticide Research.....	12
 2. LITERATURE REVIEW.....	 14
2.1 Studies on AChE Inhibitors	14
2.2 Natural vs Synthetic Pesticides	16
2.3 Molecular Docking Studies on AChE	17
 3. MATERIALS AND METHODS.....	 18
3.1 Protein Preparation (AChE – PDB ID: 6XYY)	19
3.2 Ligand Preparation (Limonene, L-carvone, Carbaryl, and Malathion).....	20
3.3 Molecular Docking Procedure.....	21
3.4 Grid Box Parameters	22
3.5 Data Analysis.....	23
3.6 ADMET Prediction Analysis.....	24
3.7 Protein–Ligand Interaction Analysis Using PLIP, PyMOL, and Discovery Studio Visualizer.....	24
 4. RESULTS	 25
4.1 Binding Energy Results	25
4.2 RMSD Analysis	27
4.3 Ki Value Comparison	28
4.4 ADMET Prediction Results	29
4.5 Active-Site Residue Interactions of the Native Ligand	32
4.6 Comparison of Amino Acid Interactions Between the Native and Docked Ligands...	34
 5. DISCUSSION	 40
5.1 Comparison of Natural and Synthetic Compounds.....	40

5.2 Interpretation of Binding Affinity, RMSD, and Ki Value.....	41
5.3 Evaluation of ADMET Prediction Results.....	41
5.4 Evaluation of Protein–Ligand Interaction Profiles.....	42
5.5 Evaluation of Natural Compounds as Alternative Pesticide Candidates.....	44
6. CONCLUSION.....	45
7. REFERENCES.....	46

LIST OF TABLES

	<u>Page</u>
Table 1. Grid box parameters used for AutoDock Vina docking simulations.....	22
Table 2. Best binding energy values of the selected ligands against AChE.....	26
Table 3. RMSD values of the selected ligands obtained from AutoDock Vina docking results.....	27
Table 4. Estimated Ki values of the selected ligands calculated from docking binding energies.....	28
Table 5. Physicochemical and SwissADME-based pharmacokinetic properties of the selected ligands.....	29
Table 6. SwissADME pharmacokinetic and drug-likeness predictions of the selected ligands.....	31
Table 7. PLIP interaction profile of the native co-crystallized ligand in AChE.....	33
Table 8. Summary of protein–ligand interactions between AChE and the native and docked ligands.....	34
Table 9. Shared residues between the native ligand and docked ligands.....	39

LIST OF FIGURES

	Page
Figure 1. Major routes of human exposure to pesticides.....	2
Figure 2. Major health effects associated with pesticide exposure.....	2
Figure 3. The role of acetylcholinesterase in normal cholinergic transmission.....	4
Figure 4. Neurotoxic consequences of organophosphate-mediated acetylcholinesterase inhibition.....	4
Figure 5. Main structural regions of acetylcholinesterase, including its active-site gorge..	6
Figure 6. Example of ligand binding within the active gorge of acetylcholinesterase.....	7
Figure 7. Major disease-related health effects associated with pesticide exposure.....	8
Figure 8. Chemical structure of limonene.....	11
Figure 9. Chemical structure of L-carvone.....	11
Figure 10. General workflow of molecular docking.....	13
Figure 11. Three-dimensional representation of acetylcholinesterase highlighting the ligand-binding region and nearby active-site residues.....	15
Figure 12. Three-dimensional structure of acetylcholinesterase obtained from PDB ID: 6XYY and visualized using PyMOL.....	20

Figure 13. Chemical structures of the ligands used in this study: limonene, L-carvone, carbaryl, and malathion.....	21
Figure 14. Docking grid box positioned around the active-site region of acetylcholinesterase and visualized using PyMOL.....	23
Figure 15. BOILED-Egg model generated by SwissADME showing the predicted gastrointestinal absorption and blood–brain barrier permeability of the selected ligands..	30
Figure 16. PLIP interaction visualization of the native co-crystallized ligand in the active-site region of AChE.....	33
Figure 17. Docking pose and two-dimensional interaction diagram of limonene in the active-site region of AChE.....	36
Figure 18. Docking pose and two-dimensional interaction diagram of L-carvone in the active-site region of AChE.....	37
Figure 19. Docking pose and two-dimensional interaction diagram of carbaryl in the active-site region of AChE.....	37
Figure 20. Docking pose and two-dimensional interaction diagram of malathion in the active-site region of AChE.....	38

LIST OF ABBREVIATIONS

AChE : Acetylcholinesterase

ADMET : Absorption, Distribution, Metabolism, Excretion, and Toxicity

BBB : Blood–Brain Barrier

CYP : Cytochrome P450

GI : Gastrointestinal

HBA : Hydrogen Bond Acceptor

HBD : Hydrogen Bond Donor

K_i : Inhibition Constant

LogP : Partition Coefficient

MW : Molecular Weight

PDB : Protein Data Bank

PDBQT : Protein Data Bank, Partial Charge, and Atom Type Format

P-gp : P-glycoprotein

PLIP : Protein–Ligand Interaction Profiler

RMSD : Root Mean Square Deviation

TPSA : Topological Polar Surface Area

Vina : AutoDock Vina

LIST OF SYMBOLS

$\text{\AA}$: Angstrom

$\text{\AA}^2$: Square Angstrom

$^{\circ}\text{C}$: Degree Celsius

kcal/mol : Kilocalorie per Mole

μM : Micromolar

$>$: Greater than

$<$: Less than

$-$: Negative Value / Minus Sign

π : Pi Interaction Symbol

INTRODUCTION

1.1 Pesticides and Their Biological Effects

Pesticides are biologically active substances used to reduce the damage caused by harmful organisms in agriculture, food storage, and public health. Their use is important because they help protect crop yield and maintain agricultural productivity. However, pesticides are not only practical agricultural chemicals; they may also interact with biological systems after exposure. For this reason, their benefits in pest control should be evaluated together with possible risks for humans, animals, and other non-target organisms (Alavanja, 2009). The toxicological importance of pesticides depends on the compound type, exposure level, and duration of contact, and both short-term and long-term effects should be considered (Mostafalou and Abdollahi, 2013).

Pesticides can be grouped according to their target organisms, such as insecticides, herbicides, fungicides, rodenticides, and fumigants. Although these groups are used for different purposes, many pesticides affect essential biological processes including enzyme activity, cellular metabolism, membrane function, or signal transmission. Therefore, their effects may extend beyond the target pest when non-target organisms are exposed directly or indirectly (Damalas and Eleftherohorinos, 2011). The biological response may vary according to dose, exposure route, frequency of contact, and the sensitivity of the exposed organism. This is especially important in occupational exposure, where repeated low-level contact may become biologically significant over time (Kim et al., 2017).

Human exposure to pesticides may occur through inhalation, dermal absorption, or ingestion of contaminated food and water. This is particularly important in agricultural settings, where workers involved in pesticide preparation and application may be repeatedly exposed during routine practice. Inadequate protective measures, improper handling, and environmental dispersion can further increase the likelihood of harmful outcomes (Damalas and Eleftherohorinos, 2011). The main routes through which humans may be exposed to pesticides are presented in Figure 1.

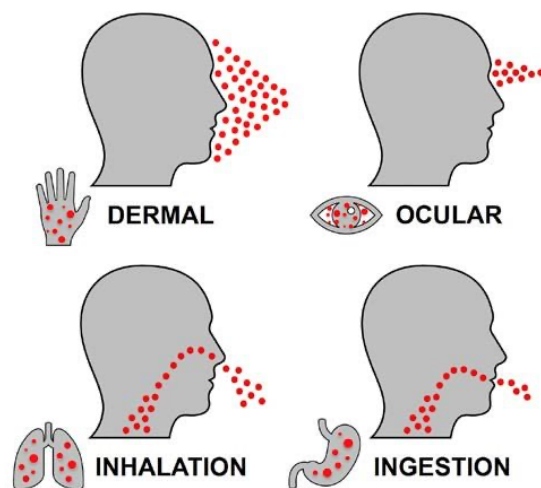


Figure 1. Major routes of human exposure to pesticides. Source: Adapted from Damalas and Eleftherohorinos (2011) and Kim et al. (2017).

After entering the body, pesticides may influence different physiological systems depending on their chemical structure, exposure level, and mechanism of action. Previous studies have reported that pesticide exposure may be related to effects on the nervous, endocrine, immune, reproductive, and developmental systems. This suggests that pesticide toxicity may involve several biological pathways rather than a single isolated target (Kim et al., 2017). The major health hazards associated with pesticide exposure are summarized in Figure 2.

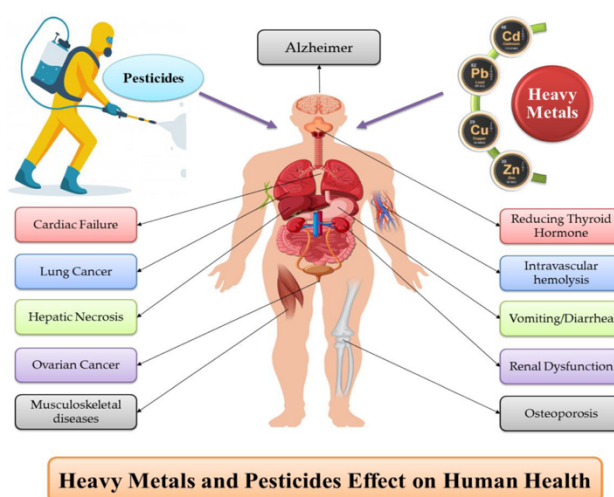


Figure 2. Major health effects associated with pesticide exposure. Source: Adapted from Alengebawy et al. (2021).

The biological effects of pesticides may be observed as acute or chronic responses. Acute exposure is generally associated with immediate symptoms of intoxication, whereas chronic exposure is more often linked to cumulative cellular injury, functional impairment, and disease development over time. This distinction is important because the toxicological significance of pesticides cannot be understood only through acute poisoning cases (Mostafalou and Abdollahi, 2013). In addition, the global scale of pesticide use increases the possibility of exposure not only for workers in direct contact with these chemicals, but also for surrounding populations and consumers exposed through environmental contamination and residue transfer. Accordingly, pesticide-related biological effects should be considered within a broader framework that includes human health, ecological balance, and long-term safety (Alavanja, 2009). Therefore, understanding pesticide mechanisms is important for balancing effective pest control with the need to reduce toxicological and environmental risks.

1.2 Neurotoxicity of Pesticides

Neurotoxicity is one of the most important adverse effects associated with pesticide exposure, particularly for compounds that are designed to act on the nervous system of target organisms. Many insecticides exert their toxic effects by interfering with neurotransmission, ion channel activity, or synaptic function, and for this reason the nervous system is considered one of the primary targets of pesticide toxicity. Since similar neurochemical pathways are also present in non-target organisms, exposure may lead to harmful neurological effects in humans and animals as well (Costa, 2006). Organophosphorus and carbamate pesticides are especially well known for their neurotoxic potential because they inhibit acetylcholinesterase, the enzyme responsible for the hydrolysis of acetylcholine at cholinergic synapses. As a result of this inhibition, acetylcholine accumulates at nerve endings, producing continuous stimulation of cholinergic receptors and disruption of normal neural transmission (Colović et al., 2013). The role of acetylcholinesterase in normal cholinergic transmission is illustrated in Figure 3.

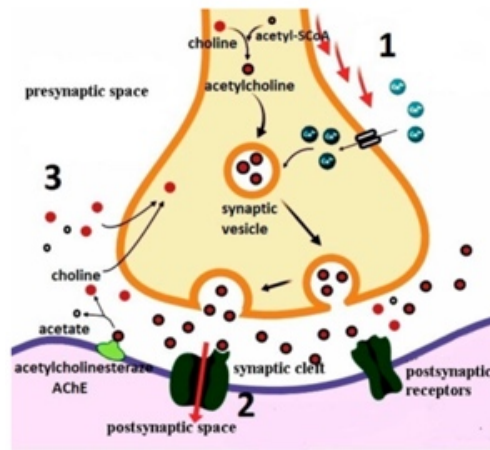


Figure 3. The role of acetylcholinesterase in normal cholinergic transmission. Source: Adapted from Stanciu et al. (2020).

The neurological effects caused by pesticide exposure may differ according to the chemical type, exposure dose, and duration of contact. In acute poisoning, excessive cholinergic stimulation may lead to symptoms such as sweating, salivation, muscle weakness, tremor, respiratory difficulty, convulsions, and in severe cases, death. These symptoms are mainly associated with the disruption of normal cholinergic signaling after AChE inhibition (Gupta, 2011). However, pesticide-induced neurotoxicity should not be limited only to acute poisoning. Repeated or long-term exposure may also contribute to persistent neurological damage through mechanisms such as oxidative stress, mitochondrial dysfunction, neuroinflammation, and altered neuronal communication (Jokanović, 2018). The downstream neurotoxic consequences of organophosphate-mediated acetylcholinesterase inhibition are illustrated in Figure 4.

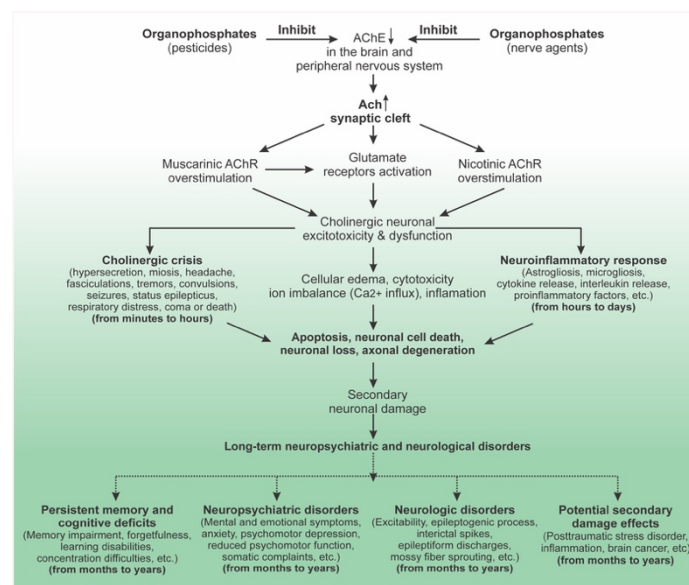


Figure 4. Neurotoxic consequences of organophosphate-mediated acetylcholinesterase inhibition. Source: Adapted from Costa (2006), Jokanović (2018).

Another important point is that pesticide neurotoxicity may occur even when exposure does not produce a dramatic poisoning event. Chronic or low-dose exposure has been associated with deficits in memory, attention, psychomotor performance, and sensory-motor function, especially in occupationally exposed populations. These observations suggest that subtle neurological alterations may develop gradually and remain clinically relevant even in the absence of severe acute symptoms (Kim et al., 2017). In addition, broader toxicological evaluations have indicated that the long-term effects of pesticide exposure may involve not only functional neurological symptoms but also progressive pathological changes linked to chronic disease processes. This makes neurotoxicity an important concern in both environmental health and preventive toxicology (Mostafalou and Abdollahi, 2013).

The mechanisms underlying pesticide-induced neurotoxicity are diverse and may differ across chemical classes. Some pesticides directly inhibit enzymes essential for neurotransmission, whereas others interfere with ion channels, membrane stability, mitochondrial activity, or intracellular signaling. Because of this mechanistic diversity, the neurotoxic profile of a pesticide cannot always be explained by a single pathway, and different compounds may produce overlapping yet distinct neurological outcomes (Costa, 2006). In the case of organophosphorus pesticides, it has also been suggested that neurotoxicity may extend beyond classical cholinergic toxicity and involve delayed or progressive neuronal injury. Such findings have strengthened the view that pesticide exposure should be evaluated not only in terms of immediate poisoning, but also in relation to long-term neurological risk (Jokanović, 2018).

Overall, pesticide-related neurotoxicity is one of the key aspects that should be considered when evaluating the biological effects of these compounds. Therefore, understanding how pesticides affect neural targets is important for interpreting their toxicological significance and for evaluating safer alternative compounds.

1.3 Acetylcholinesterase (AChE) Enzyme and Its Role

Acetylcholinesterase (AChE) is an essential enzyme for the regulation of cholinergic neurotransmission. Its main function is to hydrolyze acetylcholine after neurotransmitter release and to stop the signal at the synaptic cleft. In this way, AChE prevents prolonged stimulation of cholinergic receptors and helps maintain normal nerve impulse transmission (Colović et al., 2013). Structurally, AChE contains a deep and narrow active-site gorge, which allows the substrate and inhibitor molecules to interact with residues located inside the binding region.

This structural organization makes AChE an important enzyme for both functional nervous system regulation and inhibitor-based studies (Sussman et al., 1991). The main structural regions of acetylcholinesterase, including its active-site gorge, are illustrated in Figure 5.

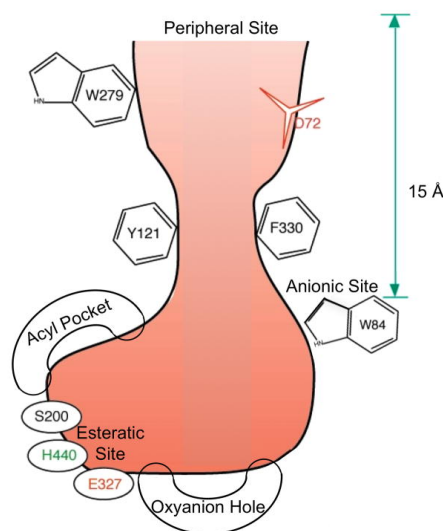


Figure 5. Main structural regions of acetylcholinesterase, including its active-site gorge. Source: Adapted from Mutunga (2011).

The physiological role of AChE is closely related to the need for balanced acetylcholine levels in the nervous system. Under normal conditions, acetylcholine is released, performs its signaling function, and is rapidly broken down by AChE. When this enzyme is inhibited, acetylcholine remains active for a longer time and continuous cholinergic stimulation may occur (Colović et al., 2013). This may affect neuromuscular transmission, autonomic nervous system activity, glandular secretion, and other processes controlled by cholinergic signaling. For this reason, AChE inhibition is a central mechanism in the toxicity of many organophosphate and carbamate pesticides (Costa, 2006). The same mechanism also explains why these compounds may be effective against insects but harmful for non-target organisms when exposure occurs (Mostafalou and Abdollahi, 2013).

The significance of AChE is not limited to pesticide toxicology alone. The enzyme also occupies an important place in pharmacology, since reversible AChE inhibition has been used therapeutically to improve cholinergic function in certain neurological conditions. This dual role makes AChE a remarkable biological target, because the same molecular system can be associated either with therapeutic benefit or with toxic damage depending on the chemical nature, reversibility, and intensity of inhibition (Colović et al., 2013). For this reason,

understanding the degree and mode of interaction between AChE and different ligands is essential when evaluating both beneficial and harmful bioactive compounds (Costa, 2006).

At the molecular level, the interaction of ligands with AChE depends on the structural features of the enzyme, particularly the residues located within and around the active-site gorge. These structural characteristics influence ligand orientation, binding stability, and inhibitory potential, which is why AChE has become a frequently studied target in structure-based analyses (Sussman et al., 1991). In addition, broader toxicological reviews have emphasized that organophosphorus compounds may affect the nervous system not only through immediate cholinergic toxicity but also through more complex pathways associated with longer-term neuronal injury (Jokanović, 2018). This further increases the importance of examining AChE interactions carefully in studies related to pesticide action. An example of ligand binding within the active gorge of acetylcholinesterase is illustrated in Figure 6.

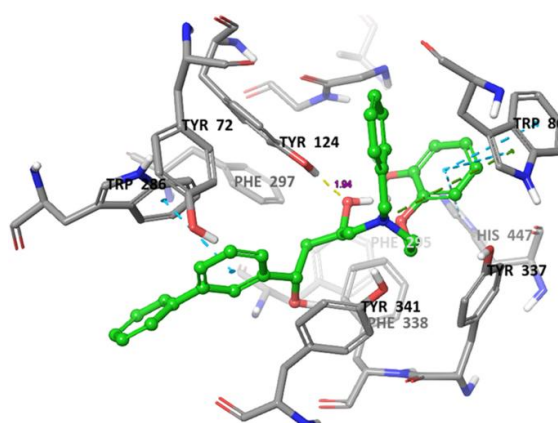


Figure 6. Example of ligand binding within the active gorge of acetylcholinesterase. Source: Adapted from Jang et al. (2018)

In pesticide-related research, AChE is a useful molecular target because ligand binding can be evaluated computationally and interpreted in relation to possible inhibitory behavior. Comparing natural and synthetic compounds against the same AChE structure allows their predicted binding affinities and active-site interaction patterns to be examined under similar conditions. Therefore, AChE-based docking analysis may provide preliminary insight into whether selected natural compounds can show meaningful interaction profiles when compared with known synthetic pesticide compounds (Mostafalou and Abdollahi, 2013).

1.4 Pesticide-Induced Diseases

Pesticide exposure has been linked with different health problems, especially in people who experience repeated occupational or environmental contact. Acute poisoning is the most visible outcome of pesticide toxicity, but it does not represent the whole health risk. Long-term or repeated exposure may also contribute to chronic pathological processes that develop gradually over time. Therefore, pesticide-related diseases should be considered not only in relation to sudden intoxication, but also in terms of cumulative biological effects (Mostafalou and Abdollahi, 2013). In addition, pesticide exposure may affect not only agricultural workers, but also nearby populations and consumers through environmental contamination, food residues, and indirect exposure routes (Kim et al., 2017). The major disease-related health effects associated with pesticide exposure are summarized in Figure 7.

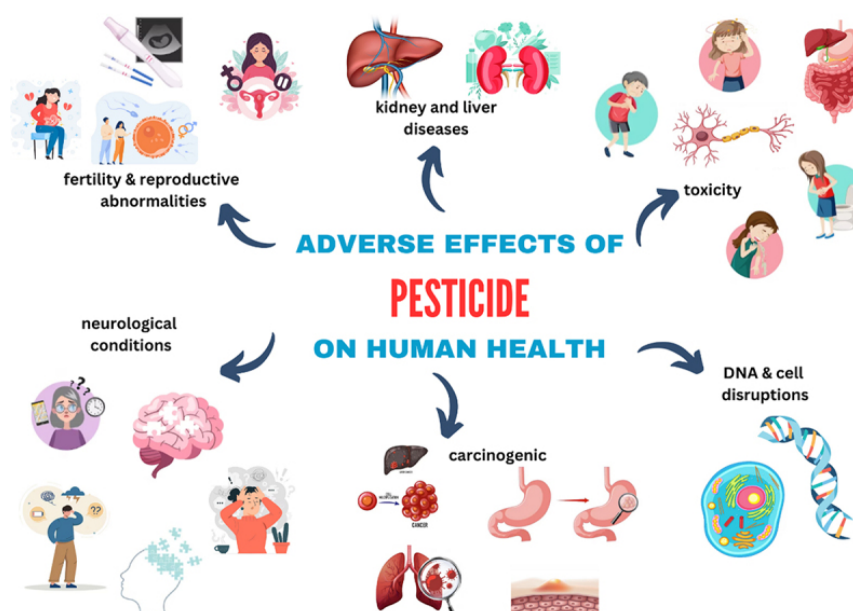


Figure 7. Major disease-related health effects associated with pesticide exposure. Source: Adapted from Mostafalou and Abdollahi (2013), Kim et al. (2017), and Alavanja and Bonner (2012).

Neurodegenerative diseases are among the most frequently discussed long-term outcomes of pesticide exposure. Parkinson's disease is an important example because several epidemiological studies have reported an association between pesticide exposure and increased disease risk. In particular, compounds such as rotenone and paraquat have been evaluated in

relation to dopaminergic neurodegeneration, supporting the idea that environmental toxicants may contribute to Parkinsonian pathology (Tanner et al., 2011). Similar epidemiological reviews have also suggested that pesticide exposure may be associated with Parkinson's disease in exposed populations, although the degree of risk may vary depending on exposure type, duration, and individual susceptibility (Freire and Koifman, 2012).

Alzheimer's disease has also been investigated in relation to pesticide exposure. Although Alzheimer's disease develops through multiple genetic, environmental, and age-related factors, occupational exposure to pesticides has been reported as a possible risk factor in elderly populations (Hayden et al., 2010). In addition to cholinergic disruption, organophosphorus pesticides may contribute to neurodegenerative processes through oxidative stress, inflammatory pathways, mitochondrial damage, and neuronal injury. These mechanisms indicate that pesticide-related neurological effects may involve more than a single toxic pathway (Mostafalou and Abdollahi, 2018).

The health effects of pesticides are not limited to classical neurodegenerative disorders. Chronic exposure has also been associated with impairments in memory, attention, psychomotor speed, and other aspects of neurological performance, particularly in occupationally exposed individuals. These findings suggest that pesticide-related neurotoxicity may develop even in the absence of severe acute poisoning and may present as more subtle but functionally important deficits (Kim et al., 2017). In the same context, broader toxicological assessments have emphasized that pesticide-related disease processes may emerge gradually and involve long-term dysfunction in multiple biological systems rather than a single isolated pathological event (Mostafalou and Abdollahi, 2013).

Beyond the nervous system, pesticide exposure has also been linked to endocrine, reproductive, and carcinogenic effects. Some pesticides have been described as endocrine-disrupting chemicals because they may interfere with hormone signaling, reproductive regulation, and developmental processes, thereby contributing to systemic health disturbances (Mnif et al., 2011). In addition, occupational exposure to pesticides has been reviewed in connection with increased cancer risk, indicating that the disease burden associated with pesticides should be considered within a broader toxicological and public health framework (Alavanja and Bonner, 2012).

Overall, pesticide exposure may be associated with both acute and chronic health effects. Neurological diseases are especially important in this context, but endocrine, reproductive, immune, and carcinogenic effects should also be considered. For this reason, understanding the

molecular targets of pesticides is important for explaining their toxic effects and for evaluating alternative compounds that may provide pest control with a lower risk profile.

1.5 Natural Compounds as Alternative Pesticides

Natural compounds have gained attention in pesticide research because of increasing concerns about the environmental persistence, non-target toxicity, and long-term health effects of some synthetic pesticides. Plant-derived substances are especially important in this area, since many of them are biodegradable and may show lower persistence in the environment compared with conventional synthetic chemicals (Isman, 2006). Essential oils and their major constituents have also been investigated as possible pest control agents because they may show insecticidal, repellent, antifungal, or growth-inhibitory effects against different pest species (Regnault-Roger et al., 2012).

The interest in natural pesticides is also related to the need for more sustainable pest management strategies. Botanical insecticides may be useful in integrated pest management because they can provide biological activity while reducing some of the problems caused by intensive synthetic pesticide use (Isman, 2006). In addition, essential oil-based compounds may act through more than one mechanism, which makes them valuable as alternative or supportive agents in agricultural pest control (Nerio et al., 2010).

Terpenes and terpenoids are among the most important groups of plant-derived pesticide candidates. They are major constituents of many essential oils and are responsible for different biological effects in insects and other pests. These compounds may affect feeding behavior, reproduction, survival, or nervous system-related functions depending on their chemical structure and target organism (Regnault-Roger et al., 2012). Their repellent activity is also important because pest management does not always require direct lethality; reducing pest attraction or feeding may also protect crops from damage (Nerio et al., 2010). The chemical structures of limonene and carvone, which are among the monoterpenes discussed in this study, are shown in Figure 8 and Figure 9.

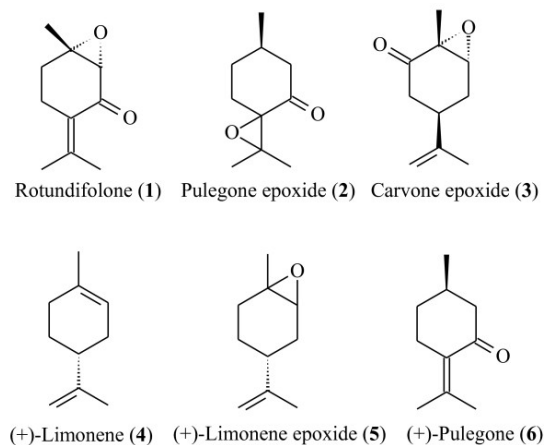


Figure 8. Chemical structure of limonene. Source: Chemical structure prepared based on PubChem Compound record.

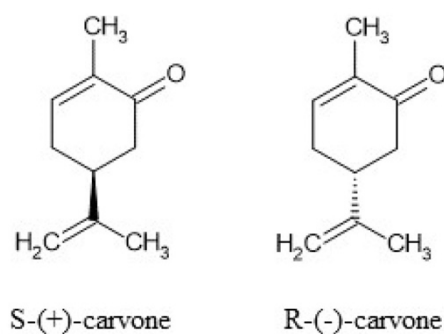


Figure 9. Chemical structure of L-carvone. Source: Chemical structure prepared based on PubChem Compound record.

One of the advantages of many natural compounds is their potential environmental compatibility. Since several plant-derived substances degrade more rapidly in soil, water, or air, they may reduce long-term residue accumulation compared with more persistent synthetic pesticides (Isman, 2006). However, natural origin should not be interpreted as automatic safety. A compound that is biologically active against pests may also affect non-target organisms under certain exposure conditions. Therefore, natural compounds should also be evaluated for efficacy, selectivity, and toxicological relevance before being considered safe alternatives (Regnault-Roger et al., 2012).

Natural pesticide candidates may also have practical limitations. Their volatility, chemical instability, variable composition, and shorter field persistence may reduce their effectiveness under agricultural conditions. These limitations mean that natural compounds may require suitable formulation strategies before they can be used reliably in pest control (Nerio et al., 2010). In addition, extraction method, plant source, storage condition, and

environmental factors may influence their biological activity. For this reason, natural pesticide candidates need detailed evaluation before practical use (Isman, 2006).

For these reasons, natural compound research should not remain limited to general biological activity screening. Understanding how these molecules interact with specific molecular targets can help explain their possible mechanisms of action. In this context, limonene and L-carvone are relevant natural monoterpenes because they can be compared with synthetic pesticide compounds at the molecular level. Their interaction with AChE may provide preliminary information about whether they show meaningful binding behavior when evaluated against a known pesticide-related enzyme target.

1.6 Molecular Docking in Drug and Pesticide Research

Molecular docking is a computational approach used to estimate how a small molecule may bind to a target protein. It predicts the possible orientation of a ligand within a selected binding region and provides an approximate binding affinity score. Because of this, docking is widely used in early-stage screening studies to compare compounds before experimental testing. It can help researchers evaluate ligand position, binding strength, and possible interaction patterns in a target protein (Meng et al., 2011). Structure-based computational methods are not limited to drug discovery; they are also useful in pesticide and toxicology research, especially when the biological target of a compound is known (Lionta et al., 2014).

One important advantage of molecular docking is that several compounds can be evaluated against the same protein under similar computational conditions. This makes it possible to compare ligands according to predicted binding energy, docking pose, and interaction profile. In this way, docking can be used as a preliminary filtering method before more expensive and time-consuming experimental studies are performed (Meng et al., 2011). However, docking results should be interpreted carefully because they are computational predictions. The reliability of a docking result depends on receptor preparation, ligand structure, scoring function, and the biological relevance of the selected binding region (Lionta et al., 2014). The general workflow of molecular docking is illustrated in Figure 10.

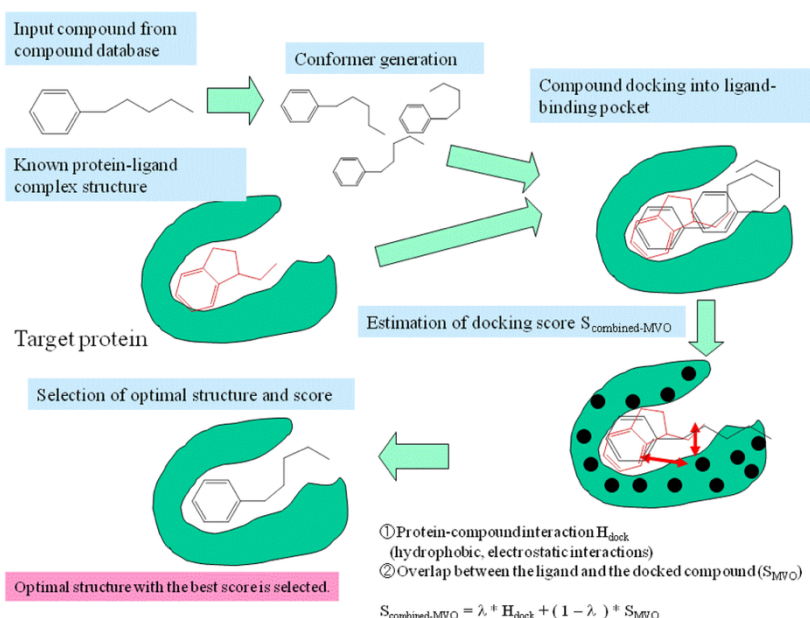


Figure 10. General workflow of molecular docking. Source: Adapted from Meng et al. (2011), Lionta et al. (2014), and Torres et al. (2019).

In pesticide-related studies, molecular docking can be useful for examining how natural or synthetic compounds interact with biological targets involved in pest survival or toxic action. AChE is one of the important targets in this context because inhibition of this enzyme is closely related to the action of several insecticides. Docking analysis can therefore provide preliminary information about whether a compound may fit into the AChE binding region and whether its interaction pattern resembles that of known inhibitors (Trott and Olson, 2010). Comparing natural monoterpenes and synthetic pesticides against the same AChE structure may also help evaluate their relative binding potential under controlled computational conditions (Lionta et al., 2014).

The quality of docking results is strongly affected by the preparation steps. The receptor structure should be cleaned properly, irrelevant molecules should be removed, and hydrogens and charges should be assigned before docking. Similarly, ligand structures should be prepared in a suitable conformation and converted into the correct file format. Grid box selection is also important because the ligand search area must include the biologically relevant binding region. If these steps are not performed carefully, the predicted binding pose and binding energy may become less reliable (Trott and Olson, 2010). Improvements in docking tools, including later versions of AutoDock Vina, have increased the usability and applicability of docking in structure-based studies (Eberhardt et al., 2021).

Docking analysis is more informative when binding scores are interpreted together with residue-level interactions. Hydrogen bonds, hydrophobic contacts, π -interactions, and other

non-covalent forces may help explain why some ligands show stronger predicted binding than others. Therefore, visual inspection and interaction analysis are important parts of docking-based interpretation. These analyses connect the numerical docking score with the possible molecular basis of ligand binding (Meng et al., 2011). For this reason, docking should be considered not only as a ranking method, but also as a tool for understanding protein–ligand recognition at the structural level (Lionta et al., 2014).

In the present study, molecular docking was used as the main computational method to compare the binding behavior of limonene, L-carvone, carbaryl, and malathion against AChE. Since these compounds differ in chemical structure and origin, evaluating them against the same target enzyme provided a controlled basis for comparison. The docking results, together with RMSD, K_i , ADMET-related predictions, and residue-level interaction analysis, allowed the selected natural and synthetic compounds to be interpreted in a more detailed way.

LITERATURE REVIEW

2.1 Studies on AChE Inhibitors

Acetylcholinesterase (AChE) inhibitors have been widely studied because AChE plays a central role in cholinergic neurotransmission. Under normal conditions, this enzyme rapidly hydrolyzes acetylcholine in the synaptic cleft and prevents prolonged stimulation of cholinergic receptors. When AChE is inhibited, acetylcholine accumulates at nerve endings and normal signal transmission may be disrupted (Colović et al., 2013).

The structure of AChE also makes it an important target for inhibitor studies. The enzyme has a narrow and deep active-site gorge where substrates and inhibitor molecules interact with specific amino acid residues. The position of a ligand inside this gorge can affect binding strength, interaction type, and possible inhibitory behavior (Sussman et al., 1991).

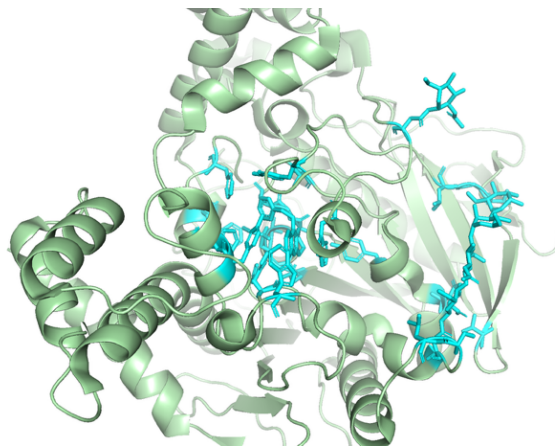


Figure 11. Three-dimensional representation of acetylcholinesterase (AChE) highlighting the ligand-binding region and nearby active-site residues. Generated from PDB ID: 6XYY using PyMOL.

AChE inhibitors are important in pharmacology because controlled inhibition of this enzyme can increase acetylcholine levels in disorders associated with reduced cholinergic signaling. However, the biological effect of an inhibitor depends not only on binding, but also on reversibility, selectivity, duration of inhibition, and safety profile (Colović et al., 2013). In toxicology, AChE inhibition is important because many organophosphate pesticides act through this mechanism. These compounds may cause acetylcholine accumulation and excessive cholinergic stimulation, leading to symptoms such as tremor, muscle weakness, respiratory difficulty, convulsions, and severe poisoning outcomes (Costa, 2006).

Carbamate pesticides are also AChE inhibitors, but their interaction with the enzyme is generally more reversible than many organophosphates. This difference is important because reversibility can influence the severity of intoxication and the recovery process. Therefore, organophosphates and carbamates are often discussed together, but their inhibitory effects should not be considered completely identical (Gupta, 2011).

The toxicological relevance of AChE inhibition is not limited to acute poisoning. Long-term or repeated exposure to some pesticide classes has also been associated with delayed neurological effects. These effects may involve oxidative stress, mitochondrial dysfunction, inflammation, and disruption of neuronal signaling pathways (Jokanović, 2018).

Natural products have also become important in AChE inhibitor research. Several plant-derived compounds, including alkaloids, coumarins, terpenes, and polyphenols, have been reported to show AChE inhibitory activity at different levels. This suggests that natural molecules may provide alternative structures for screening studies and may help identify compounds with potential biological activity (dos Santos et al., 2018).

Terpene-based compounds are especially relevant for pesticide research because many of them are major components of essential oils. Essential oils and their constituents may show insecticidal, repellent, antifeedant, or growth-inhibitory effects against different pest species. Since some of these compounds may also affect neural targets in insects, their possible interaction with AChE should be evaluated at the molecular level (Regnault-Roger et al., 2012).

In this study, natural monoterpenes such as limonene and L-carvone were compared with synthetic pesticides such as carbaryl and malathion using molecular docking. This

comparison allowed all selected ligands to be evaluated against the same AChE structure. Therefore, binding affinity values, docking poses, and active-site residue interactions could be interpreted together to compare the predicted inhibitory potential of natural and synthetic compounds.

2.2 Natural vs Synthetic Pesticides

Synthetic pesticides have an important role in modern agriculture because they provide rapid and effective control against insects, weeds, fungi, and other harmful organisms. Their use helps protect crop yield, reduce production losses, and support agricultural productivity. However, repeated and widespread pesticide use may also create concerns related to environmental contamination, non-target toxicity, and long-term health effects. For this reason, synthetic pesticides should be evaluated not only according to their effectiveness, but also according to their possible biological and environmental impacts (Alavanja, 2009).

Many synthetic pesticides affect essential biological processes in target organisms. Organophosphate and carbamate insecticides are important examples because they inhibit acetylcholinesterase and interfere with nervous system function. Although this mechanism is useful for pest control, it may also create toxic effects in non-target organisms because similar cholinergic pathways are found in humans and animals (Costa, 2006).

The toxicological risk of synthetic pesticides depends on their chemical structure, dose, exposure route, persistence, and mechanism of action. Some compounds may cause acute poisoning after high exposure, while others may contribute to chronic health problems after repeated contact. Therefore, pesticide safety cannot be judged only by short-term agricultural efficiency (Mostafalou and Abdollahi, 2013). Environmental persistence is also important because residues may remain in soil, water, or food systems and may increase indirect exposure risks for workers, consumers, and ecosystems (Damalas and Eleftherohorinos, 2011).

Because of these concerns, natural pesticides have gained attention as possible alternatives or supportive agents in pest management. Natural pesticides are generally derived from plants, microorganisms, or other biological sources. Plant-derived compounds are especially valuable because many of them are biodegradable and may show lower environmental persistence than conventional synthetic pesticides (Isman, 2006). Botanical insecticides and essential oil constituents may show insecticidal, repellent, antifeedant, or growth-regulating effects, which makes them useful candidates for integrated pest management (Regnault-Roger et al., 2012).

Essential oils contain many bioactive compounds, especially terpenes and terpenoids. Their biological effects may result from the combined activity of different molecules rather than a single compound. This can be an advantage because it may provide multiple modes of action, but it may also create problems in standardization, formulation, and reproducibility (Nerio et al., 2010). In addition, natural origin does not automatically mean complete safety. A natural compound that affects pests may also affect non-target organisms depending on dose, exposure route, and chemical properties. Therefore, natural compounds should also be

evaluated carefully in terms of efficacy, selectivity, and toxicological profile (Regnault-Roger et al., 2012).

Compared with many synthetic pesticides, natural compounds may have practical limitations such as higher volatility, lower stability, variable composition, and shorter field activity. These features may reduce their effectiveness under agricultural conditions unless appropriate formulations are developed (Isman, 2006). Still, natural compounds remain important because they may provide new chemical structures for pesticide research. Instead of directly replacing all synthetic pesticides, they may be used as lead compounds, complementary agents, or safer alternatives in specific pest control strategies (Nerio et al., 2010).

For this reason, the comparison between natural and synthetic pesticides should not be reduced to a simple “safe versus harmful” distinction. A more accurate comparison should include molecular target, binding behavior, biological activity, persistence, selectivity, and toxicological risk. Computational methods can support this comparison by evaluating different compounds under the same target-based conditions (Meng et al., 2011).

In the present study, limonene and L-carvone were not assumed to be automatically safer or more effective because they are natural compounds. Instead, they were compared with carbaryl and malathion through molecular docking, binding affinity values, and amino acid interaction profiles. This approach allowed the selected natural and synthetic compounds to be evaluated more objectively according to their predicted interactions with the AChE active-site region.

2.3 Molecular Docking Studies on AChE

Molecular docking studies on acetylcholinesterase are widely used because AChE has a well-defined binding region and is directly related to both pesticide toxicity and enzyme inhibition studies. Docking allows the possible placement of small molecules inside the AChE active-site gorge to be predicted before experimental testing. This makes it useful for comparing different inhibitor candidates and for understanding how ligands may interact with important amino acid residues in the enzyme (Meng et al., 2011). Since the active-site gorge of AChE is narrow and structurally specific, ligand orientation inside this region may strongly affect predicted binding stability and inhibitory potential (Sussman et al., 1991).

In docking studies, binding affinity is generally used as the first numerical parameter for comparing ligand–protein interaction strength. A lower binding energy value usually suggests a more favorable predicted interaction. However, docking scores should not be interpreted alone because they are affected by receptor preparation, ligand conformation, grid box selection, and the scoring function used by the software (Trott and Olson, 2010). For this reason, residue-level interaction analysis is also important. Hydrogen bonds, hydrophobic contacts, π – π interactions, π –cation interactions, and other non-covalent interactions can help explain why one ligand may show stronger or weaker binding than another (Lionta et al., 2014).

AChE docking studies have been used for both synthetic inhibitors and natural product candidates. Synthetic pesticides and known inhibitors provide useful reference points because many of them are already associated with AChE inhibition. Natural compounds, on the other

hand, are valuable for screening because they may provide alternative chemical structures with possible biological activity. Evaluating both groups under the same computational conditions makes the comparison more consistent (Colović et al., 2013). Several plant-derived compounds, including terpenes, alkaloids, coumarins, and polyphenols, have been reported as possible AChE inhibitors, which supports the use of docking as a preliminary screening method for natural molecules (dos Santos et al., 2018).

In pesticide-related docking research, AChE is especially important because organophosphate and carbamate pesticides act mainly by interfering with cholinergic neurotransmission. Their inhibitory effect on AChE is closely connected to their toxic action. Therefore, comparing natural compounds with synthetic pesticides on the same AChE structure can help show whether natural molecules have similar, weaker, or different binding behavior (Costa, 2006). To make this comparison reliable, the receptor and ligands must be prepared carefully. The protein should be cleaned, unnecessary molecules should be removed, hydrogens and charges should be added, and the docking region should be selected according to the biologically relevant binding site. Ligands should also be prepared in suitable conformations before docking (Trott and Olson, 2010).

AutoDock Vina is commonly used in docking studies because it provides relatively fast ligand screening and reports predicted binding affinity values. It is suitable for comparing small molecules against protein targets and has been widely used in structure-based studies. Later versions of Vina improved different parts of the docking procedure, including force-field options and usability, which increased its application in computational screening studies (Eberhardt et al., 2021).

Docking results become more meaningful when numerical values are supported by pose inspection and interaction analysis. RMSD values can help evaluate the variation between predicted poses, while molecular visualization shows whether the ligand is positioned in the expected binding region. In addition, interaction tools can identify the residues involved in ligand binding and describe interaction types such as hydrogen bonds, hydrophobic interactions, salt bridges, and aromatic contacts (Meng et al., 2011). PLIP is one of the tools used for this purpose because it automatically detects protein–ligand interactions and supports residue-level interpretation of docking results (Salentin et al., 2015).

In the present study, molecular docking was used to compare limonene, L-carvone, carbaryl, and malathion against AChE. The comparison was not based only on binding energy values. RMSD values, estimated K_i values, ADMET-related predictions, and protein–ligand interaction profiles were also evaluated. This approach made it possible to compare the selected natural and synthetic compounds more clearly in terms of their predicted AChE binding behavior.

3. MATERIALS AND METHODS

In this study, a computational molecular docking workflow was used to compare the interaction of selected natural and synthetic pesticide-related compounds with acetylcholinesterase (AChE). Limonene and L-carvone were selected as natural monoterpenes, while carbaryl and malathion were selected as synthetic pesticide compounds.

All compounds were evaluated against the same AChE structure to make the comparison more consistent.

The AChE structure was obtained from the Protein Data Bank using the PDB ID 6XYY. The chemical structures of limonene, L-carvone, carbaryl, and malathion were obtained from PubChem. Before docking, the receptor and ligand structures were prepared by cleaning the protein, checking ligand structures, assigning the required hydrogens and charges, and converting the files into PDBQT format.

Docking simulations were carried out using AutoDock Vina. The docking grid was positioned around the AChE ligand-binding region, and the same grid parameters were used for all ligands. The results were evaluated according to binding affinity, RMSD values, estimated K_i values, and docking pose positions.

After docking, the ligand poses were visualized using PyMOL. Protein–ligand interactions were examined using PLIP to identify residue-level contacts such as hydrophobic interactions, hydrogen bonds, π -stacking, π -cation interactions, water bridges, and salt bridges. Discovery Studio Visualizer was also used to prepare two-dimensional interaction diagrams for the docked ligands.

The native co-crystallized ligand of the AChE structure was analyzed separately as a reference. Its interaction residues were compared with the interaction profiles of limonene, L-carvone, carbaryl, and malathion. In this way, the ligands were evaluated not only by docking scores, but also by their residue overlap and interaction patterns within the AChE binding region.

3.1 Protein Preparation (AChE – PDB ID: 6XYY)

The three-dimensional structure of acetylcholinesterase was downloaded from the Protein Data Bank using the PDB ID 6XYY. This structure was selected because it contains a defined ligand-binding region suitable for docking analysis. The protein file was first examined in PyMOL to observe the general structure, protein chain, native ligand position, and active-site region.

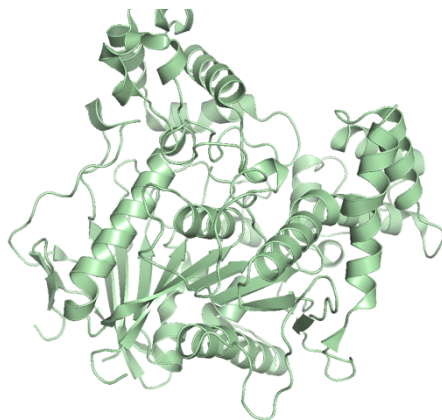


Figure 12. Three-dimensional structure of acetylcholinesterase obtained from PDB ID: 6XYY and visualized using PyMOL.

Before docking, the receptor structure was prepared by removing water molecules and non-essential components from the protein file. The native co-crystallized ligand was also removed from the receptor used in docking simulations. However, this native ligand was not discarded completely; it was analyzed separately later to identify reference active-site residues for comparison with the docked ligands.

After cleaning, polar hydrogens and required charges were added to the AChE structure. The prepared receptor was then converted into PDBQT format for AutoDock Vina. The same receptor file was used for all docking simulations so that limonene, L-carvone, carbaryl, and malathion could be compared under identical protein preparation conditions.

3.2 Ligand Preparation (Limonene, L-carvone, Carbaryl, and Malathion)

Four ligands were selected for the molecular docking analysis in this study. Limonene and L-carvone were included as natural monoterpene compounds, while carbaryl and malathion were selected as representative synthetic pesticide compounds. This selection was made to compare natural and synthetic pesticide-related molecules against the same acetylcholinesterase target under similar computational conditions.

The chemical structures of limonene, L-carvone, carbaryl, and malathion were obtained from the PubChem database. Before docking, the structures were checked to confirm that the correct compounds were used in the analysis. This step was important because differences in molecular structure, stereochemistry, functional groups, and atomic arrangement may influence ligand conformation, docking pose, and predicted binding affinity.

Before molecular docking, all ligand structures were prepared using the same general procedure. The ligand files were first arranged in a suitable structural format and then energy minimization was performed to obtain more stable conformations before docking. Energy minimization was included to reduce unfavorable geometry and to make the ligand structures more appropriate for docking simulation.

After preparation, each ligand was converted into PDBQT format, which is required for AutoDock Vina docking calculations. The same ligand preparation procedure was applied to limonene, L-carvone, carbaryl, and malathion in order to reduce methodological differences between the natural and synthetic compounds. The prepared ligand files were then used as input structures for docking simulations against the prepared AChE receptor.

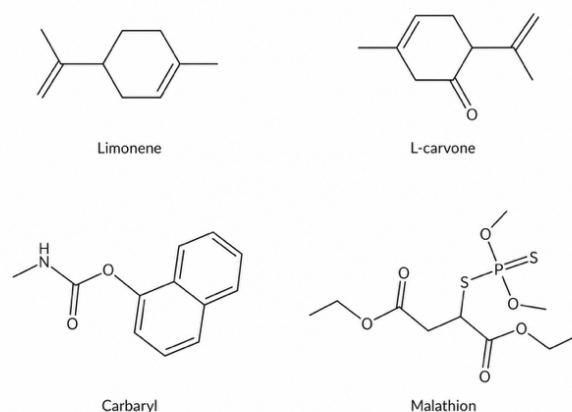


Figure 13. Chemical structures of the ligands used in this study: limonene, L-carvone, carbaryl, and malathion.

3.3 Molecular Docking Procedure

Molecular docking simulations were performed using AutoDock Vina to evaluate the possible binding modes of limonene, L-carvone, carbaryl, and malathion within the ligand-binding region of acetylcholinesterase. The prepared AChE receptor file and the prepared ligand files were used in PDBQT format, which is the required input format for AutoDock Vina.

Each ligand was docked separately against the same prepared AChE receptor. The use of the same receptor structure for all docking simulations was important because it allowed the selected ligands to be compared under identical protein conditions. In this way, differences observed between ligands were interpreted as ligand-related differences rather than differences caused by receptor preparation.

During the docking procedure, AutoDock Vina generated multiple binding conformations, also called docking poses, for each ligand. These poses were ranked according to their predicted binding affinity values, expressed in kcal/mol. For each compound, the pose with the lowest binding energy was accepted as the best predicted binding mode. A more negative binding energy value was interpreted as a more favorable predicted ligand–receptor interaction.

However, the docking score was not used as the only evaluation criterion. The best-ranked docking poses were also examined visually using PyMOL to confirm whether the ligands were positioned within the expected AChE binding region. This visual inspection was

important because a favorable docking score alone may not be sufficient if the ligand is not located in a biologically meaningful binding area.

The docking outputs obtained from AutoDock Vina were recorded for all four ligands. Binding affinity values and RMSD values were used as the first stage of comparison. The results were then evaluated together with estimated K_i values and protein–ligand interaction analysis. This combined evaluation allowed the predicted inhibitory potential of the selected natural and synthetic compounds to be interpreted more carefully.

3.4 Grid Box Parameters

The docking grid box was arranged around the ligand-binding region of acetylcholinesterase in order to evaluate the possible binding positions of the selected compounds within a biologically relevant area. Since the aim of this study was to compare limonene, L-carvone, carbaryl, and malathion under the same docking conditions, the same grid box parameters were used for all ligands.

The grid center was selected according to the active-site region of the AChE structure. This region was chosen because it contains residues that may contribute to ligand binding and enzymatic inhibition. Defining the grid around this area allowed AutoDock Vina to search for possible ligand conformations within or near the AChE binding pocket rather than across the whole protein surface.

The grid box dimensions were kept large enough to include the ligand-binding pocket and surrounding amino acid residues that could participate in protein–ligand interactions. Using identical grid parameters for all docking simulations was important for reducing methodological variation between ligands. Therefore, the differences observed in binding energy, RMSD, and interaction profiles were interpreted as compound-related differences rather than differences caused by grid placement.

The grid center and size values used in AutoDock Vina are shown in Table 1.

Table 1. Grid box parameters used for AutoDock Vina docking simulations.

Parameter	Value
Center X	24.119
Center Y	64.581
Center Z	9.023
Size X	30
Size Y	30
Size Z	30
Grid spacing	0.375 Å

The grid spacing was set to 0.375 Å, and the grid box dimensions were kept constant for all ligands. This allowed the docking analysis to focus on differences in ligand binding behavior rather than differences caused by grid placement or docking area selection.

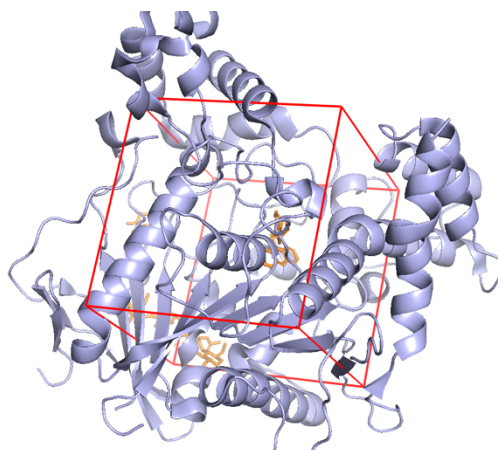


Figure 14. Docking grid box positioned around the active-site region of acetylcholinesterase and visualized using PyMOL.

3.5 Data Analysis (Binding Affinity, RMSD, K_i , ADMET and Active-Site Residue Interaction Analysis)

The docking outputs were evaluated first by using the binding affinity values obtained from AutoDock Vina. The values were reported in kcal/mol, and the most negative binding energy was considered the most favorable predicted ligand–receptor interaction. For each ligand, the best docking pose was selected according to the lowest binding energy value.

RMSD values were also examined to evaluate the variation between the docking poses generated for each ligand. The RMSD values reported by AutoDock Vina were used to compare how close the alternative poses were to the best-ranked pose. Lower RMSD values indicated closer pose similarity, whereas higher RMSD values suggested greater conformational variation among the predicted docking modes.

In addition to binding affinity and RMSD values, estimated inhibition constant values (K_i) were calculated from the binding energy results. K_i values were used as a supportive numerical parameter for comparing the predicted inhibitory potential of the selected ligands. In general, lower K_i values were interpreted as stronger predicted inhibition, while higher K_i values indicated weaker predicted inhibition.

The numerical docking results were interpreted together with ADMET-related predictions and protein–ligand interaction profiles. This was important because binding energy alone may not fully explain the biological relevance of a predicted docking pose. Therefore, binding affinity, RMSD, estimated K_i values, ADMET-related properties, and active-site residue interactions were evaluated together to compare the predicted binding behavior of limonene, L-carvone, carbaryl, and malathion.

The native co-crystallized ligand was used as a reference for identifying important residues in the AChE binding region. The amino acid residues interacting with the native ligand were compared with those interacting with the docked ligands. This comparison helped determine whether the selected ligands shared similar residue-level interaction patterns with the native ligand.

3.6 ADMET Prediction Analysis

Selected physicochemical and ADMET-related properties of limonene, L-carvone, carbaryl, and malathion were evaluated using *in silico* prediction-based analysis. Molecular weight, consensus LogP, topological polar surface area, hydrogen bond donor number, hydrogen bond acceptor number, rotatable bond number, and Lipinski rule violations were used to compare the basic molecular properties of the selected ligands.

In addition to these descriptors, pharmacokinetic-related predictions were evaluated to support the ligand comparison. These included gastrointestinal absorption, blood–brain barrier permeability, P-glycoprotein substrate prediction, CYP inhibition profile, Lipinski rule evaluation, and selected drug-likeness parameters. These results were interpreted only as computational screening data and were not considered equivalent to experimental pharmacokinetic or toxicity findings.

The ADMET-related predictions were used together with docking and interaction results. This allowed the selected natural and synthetic compounds to be compared not only according to binding affinity and active-site residue interactions, but also according to predicted physicochemical and pharmacokinetic-related profiles.

3.7 Protein–Ligand Interaction Analysis Using PLIP, PyMOL, and Discovery Studio Visualizer

Protein–ligand interaction analysis was performed to examine the amino acid residues involved in ligand binding and to interpret the docking results at the residue level. For this purpose, PyMOL, PLIP, and Discovery Studio Visualizer were used after molecular docking simulations.

PyMOL was used for visual inspection of docking poses and preparation of three-dimensional molecular figures. PLIP was used to identify residue-level interaction types between the receptor and ligands, including hydrophobic interactions, hydrogen bonds, π -stacking interactions, π -cation interactions, salt bridges, and water bridges. For each docked ligand, the corresponding protein–ligand complex was prepared and analyzed to record the interacting amino acid residues.

The native co-crystallized ligand of the AChE structure was also analyzed separately. This step was included to identify the key residues located in the original ligand-binding region of AChE. The interaction profile of the native ligand was then used as a reference for evaluating the docked ligands.

After PLIP analysis, the residues interacting with limonene, L-carvone, carbaryl, and malathion were compared with the residues interacting with the native ligand. Shared residues were considered important because residue overlap may support the relevance of a predicted docking pose within the AChE binding region.

Discovery Studio Visualizer was used to generate two-dimensional protein–ligand interaction diagrams for the docked ligands. These diagrams were used as visual support for interaction types such as van der Waals contacts, alkyl interactions, π -alkyl interactions, conventional hydrogen bonds, and pi-sulfur interactions.

The final comparison was based on both numerical docking outputs and residue-level interaction profiles. Ligands with lower binding energy values, lower estimated K_i values, and interaction overlap with the native ligand were interpreted as having stronger and more relevant predicted binding behavior toward AChE.

4. RESULTS

In this study, molecular docking analysis was performed to compare the binding behavior of two natural monoterpenes, limonene and L-carvone, with two synthetic pesticide compounds, carbaryl and malathion, against acetylcholinesterase (AChE). The results were evaluated using binding affinity values, RMSD values, estimated inhibition constant (K_i) values, ADMET-related prediction results, and protein–ligand interaction profiles obtained from PLIP and PyMOL analysis.

The results were interpreted using both numerical docking outputs and residue-level interaction data. Binding energy, RMSD, and K_i values were first examined to compare the predicted binding strength and pose consistency of the ligands. Then, ADMET-related properties were evaluated to support the comparison of the selected natural and synthetic compounds in terms of predicted physicochemical and pharmacokinetic-related profiles. Finally, the amino acid interaction profiles of the docked ligands were compared with the interaction profile of the native co-crystallized ligand of AChE.

4.1 Binding Energy Results

Binding affinity values obtained from AutoDock Vina were used as the first parameter to compare the predicted binding behavior of limonene, L-carvone, carbaryl, and malathion against AChE. Binding energy values were expressed in kcal/mol. In docking analysis, a more

negative binding energy value generally indicates a more favorable predicted interaction between the ligand and the receptor.

The best binding energy values obtained for the selected ligands are shown in Table 2. Among all tested compounds, carbaryl showed the lowest binding energy value with -8.489 kcal/mol, indicating the strongest predicted binding affinity toward AChE. L-carvone showed the strongest binding among the natural compounds with -6.675 kcal/mol, while limonene showed a slightly weaker binding value of -6.356 kcal/mol. Malathion showed a binding energy of -6.611 kcal/mol, which was close to the value obtained for L-carvone.

Table 2. Best binding energy values of the selected ligands against AChE

Ligand	Compound type	Best binding energy (kcal/mol)
Limonene	Natural monoterpene	-6.356
L-carvone	Natural monoterpene	-6.675
Carbaryl	Synthetic pesticide	-8.489
Malathion	Synthetic pesticide	-6.611

Based on the binding energy results, the ligands were ranked from strongest to weakest predicted binding as follows: carbaryl > L-carvone > malathion > limonene. Carbaryl was clearly separated from the other compounds by its more negative binding energy value. In contrast, L-carvone and malathion showed relatively close binding affinities, suggesting that the natural compound L-carvone had a comparable predicted binding potential to malathion under the docking conditions used in this study.

Although limonene showed the weakest binding energy among the tested ligands, its value still indicated a possible interaction with the AChE binding region. Therefore, the binding energy results suggested that all selected compounds were able to form predicted interactions with AChE, but their binding strengths differed. Overall, carbaryl showed the most favorable docking score, while L-carvone showed the most favorable binding result among the natural monoterpenes.

4.2 RMSD Analysis

RMSD values were evaluated in order to examine the variation between the predicted docking poses generated by AutoDock Vina. In the docking output, the first mode represents the best-ranked pose with the lowest binding energy and is therefore reported with an RMSD value of 0.000 Å. The following modes were compared with the best pose by using lower-bound RMSD values.

The RMSD results of the second and third docking modes are shown in Table 3. Limonene showed RMSD values of 1.028 Å for mode 2 and 1.720 Å for mode 3, indicating that its alternative poses were relatively close to the best docking pose. Carbaryl also showed consistent alternative poses, with RMSD values of 1.611 Å and 1.549 Å for modes 2 and 3, respectively. These values suggest that the predicted docking conformations of carbaryl were relatively stable around the best binding mode.

Table 3. RMSD values of the selected ligands obtained from AutoDock Vina docking results.

Ligand	Best binding energy (kcal/mol)	Mode 2 RMSD (Å)	Mode 3 RMSD (Å)
Limonene	−6.356	1.028	1.720
L-carvone	−6.675	2.186	1.418
Malathion	−6.611	3.037	1.352
Carbaryl	−8.489	1.611	1.549

L-carvone showed an RMSD value of 2.186 Å for mode 2 and 1.418 Å for mode 3. Although the second mode was slightly above 2 Å, the third mode was closer to the best pose, suggesting acceptable pose consistency. Malathion showed a higher RMSD value of 3.037 Å for mode 2, while mode 3 had a lower RMSD value of 1.352 Å. This indicates that one of the alternative malathion poses was farther from the best binding mode, but another predicted pose remained closer to the best-ranked conformation.

Overall, the RMSD analysis showed that limonene and carbaryl had relatively consistent alternative docking poses, while L-carvone and malathion showed greater variation in at least one alternative mode. However, the presence of lower RMSD values in the following modes

suggests that the docking results were generally acceptable for comparative interpretation. Therefore, RMSD analysis supported the use of the best docking poses for further K_i calculation and interaction analysis.

4.3 K_i Value Comparison

Estimated inhibition constant values (K_i) were calculated from the best binding energy values in order to compare the predicted inhibitory potential of the selected ligands. K_i values provide an additional numerical parameter for interpreting docking results. In general, a lower K_i value indicates stronger predicted inhibition, while a higher K_i value indicates weaker predicted inhibition.

The calculated K_i values of limonene, L-carvone, malathion, and carbaryl are shown in Table 4. Carbaryl had the lowest K_i value with 0.60 μM , indicating the strongest predicted inhibitory potential among the tested ligands. L-carvone showed a K_i value of 12.88 μM , while malathion showed a similar value of 14.16 μM . Limonene had the highest K_i value with 21.79 μM , indicating the weakest predicted inhibitory potential among the four compounds.

Table 4. Estimated K_i values of the selected ligands calculated from docking binding energies.

Ligand	Best binding energy ΔG (kcal/mol)	Estimated K_i (μM)
Limonene	-6.356	21.79
L-carvone	-6.675	12.88
Malathion	-6.611	14.16
Carbaryl	-8.489	0.60

The K_i results followed the same general trend as the binding energy results. Carbaryl showed the most favorable predicted inhibition profile, which was consistent with its lowest binding energy value. Among the natural compounds, L-carvone had a lower K_i value than limonene, suggesting stronger predicted inhibitory potential. Malathion and L-carvone showed

relatively close K_i values, supporting the idea that L-carvone may have a comparable predicted inhibitory profile to malathion under the computational conditions used in this study.

Overall, the K_i comparison supported the binding energy results. Carbaryl appeared as the strongest predicted AChE inhibitor among the tested ligands, while L-carvone showed the most favorable result among the natural monoterpenes. Limonene showed the weakest predicted inhibitory potential, although its K_i value still indicated measurable predicted interaction with AChE.

4.4 ADMET Prediction Results

In addition to molecular docking results, selected ADMET-related properties were evaluated in order to support the comparison of limonene, L-carvone, carbaryl, and malathion. These parameters were used as in silico prediction-based screening data and were not considered equivalent to experimental pharmacokinetic or toxicity results. The physicochemical descriptor results are shown in Table 5.

Table 5. Physicochemical and SwissADME-based pharmacokinetic properties of the selected ligands.

Ligand	MW (g/mol)	Conse nsus LogP	TPSA (Å ²)	HB D	HB A	Rot B	Lipinsk i violatio n	GI absorpti on	BBB permean t
Limonene	136.23	3.37	0.00	0	0	1	0	Low	Yes
L-carvone	150.22	2.44	17.07	0	1	1	0	High	Yes
Carbaryl	201.22	2.32	38.33	1	2	3	0	High	Yes
Malathion	330.36	2.14	138.26	0	6	11	0	Low	No

All selected ligands showed zero Lipinski rule violations, suggesting that their basic physicochemical properties were within commonly accepted ranges for small-molecule

screening. Limonene had the lowest molecular weight and TPSA value, which is consistent with its small and highly hydrophobic structure. L-carvone also showed a low molecular weight and low TPSA value, while carbaryl had intermediate physicochemical properties. Malathion had the highest molecular weight, TPSA, and rotatable bond number among the selected compounds.

The SwissADME results showed that L-carvone and carbaryl had high predicted gastrointestinal absorption, while limonene and malathion showed low predicted gastrointestinal absorption. Limonene, L-carvone, and carbaryl were predicted to be blood–brain barrier permeant, whereas malathion was not predicted to cross the blood–brain barrier. The BOILED-Egg model was used as a visual support for interpreting the predicted gastrointestinal absorption and blood–brain barrier permeability of the selected ligands.

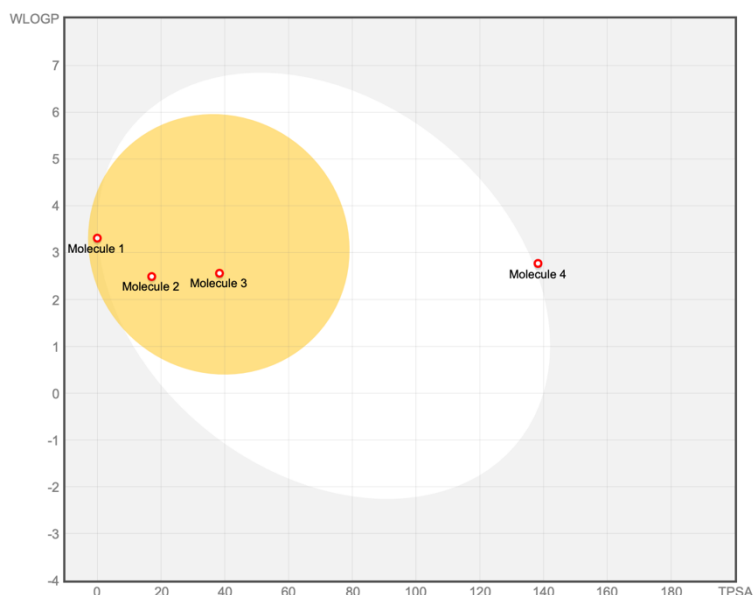


Figure 15. BOILED-Egg model generated by SwissADME showing the predicted gastrointestinal absorption and blood–brain barrier permeability of the selected ligands. Molecule 1: limonene; Molecule 2: L-carvone; Molecule 3: carbaryl; Molecule 4: malathion.

In the BOILED-Egg model, limonene, L-carvone, and carbaryl were located within the yellow region, suggesting predicted blood–brain barrier permeability. Malathion was positioned outside the main BOILED-Egg region and was not predicted to be BBB permeant. This result was consistent with its higher TPSA and higher number of rotatable bonds compared with the other ligands. Therefore, although malathion showed a docking affinity close to L-carvone, its

BOILED-Egg position indicates a different predicted pharmacokinetic-related profile rather than simply a weaker result.

The different position of malathion in the BOILED-Egg plot is especially important because it reflects its distinct physicochemical profile compared with the other ligands. Limonene, L-carvone, and carbaryl were located on the left side of the plot with lower TPSA values, indicating lower molecular polarity and a higher probability of passive membrane permeability. In contrast, malathion was located on the right side of the plot due to its markedly higher TPSA value and higher number of rotatable bonds. This suggests that malathion is more polar and structurally more flexible than the other compounds, which may reduce its passive gastrointestinal absorption and prevent blood–brain barrier permeability. Therefore, although malathion showed a docking affinity close to L-carvone, its BOILED-Egg position indicates a different predicted pharmacokinetic-related profile rather than simply a weaker or less favorable result. This difference supports the importance of evaluating docking results together with ADMET-related predictions rather than interpreting binding energy alone. These ADMET findings were used only as supportive *in silico* screening data and were interpreted together with docking results and protein–ligand interaction profiles.

Additional SwissADME pharmacokinetic and drug-likeness predictions are summarized in Table 6.

Table 6. SwissADME pharmacokinetic and drug-likeness predictions of the selected ligands.

Ligand	P-gp substrate	CYP inhibition profile	Lipinski	Additional drug-likeness notes
Limonene	No	CYP2C9 inhibitor: Yes; others: No	0 violation	Low MW-related rule alerts
L-carvone	No	No major CYP inhibition predicted	0 violation	Low MW-related rule alerts
Carbaryl	No	CYP1A2 inhibitor: Yes; others: No	0 violation	Low MW-related rule alert
Malathion	No	CYP2C19 inhibitor: Yes; others: No	0 violation	Veber violation due to rotatable bonds >10; Egan violation due to TPSA >131.6

The SwissADME pharmacokinetic predictions showed that none of the selected ligands were predicted to be P-gp substrates. L-carvone showed the simplest CYP profile among the tested compounds, with no major CYP inhibition predicted. Limonene was predicted as a CYP2C9 inhibitor, while carbaryl was predicted as a CYP1A2 inhibitor. Malathion was predicted as a CYP2C19 inhibitor and also showed additional rule violations related to high rotatable bond number and high TPSA.

Overall, the ADMET prediction results suggested that the natural monoterpenes and carbaryl had more favorable passive BBB-related profiles than malathion. However, limonene showed low predicted GI absorption despite its BBB permeability prediction. Malathion showed the most distinct SwissADME profile in this comparison because it had low predicted GI absorption, no predicted BBB permeability, high TPSA, and a high number of rotatable bonds.

4.5 Active-Site Residue Interactions of the Native Ligand

The native co-crystallized ligand of the AChE structure was analyzed in order to identify the key residues involved in the original ligand-binding region. This analysis was important because the interaction profile of the native ligand provided a reference point for evaluating the docked ligands. The native ligand was identified as 760-A-602 in the selected AChE structure. PLIP analysis showed that the native ligand interacted with several residues located in or near the AChE binding pocket. The main interaction types observed for the native ligand were hydrophobic interactions, water bridges, π -stacking, and π -cation interactions. The residues involved in these interactions are shown in Table 7.

Table 7. PLIP interaction profile of the native co-crystallized ligand in AChE.

Residue	Interaction type
Tyr71	Hydrophobic interaction
Trp83	Hydrophobic interaction, water bridge, π -stacking
Phe330	Hydrophobic interaction
Tyr370	Hydrophobic interaction
Phe371	Hydrophobic interaction
Trp472	Hydrophobic interaction
Thr154	Water bridge
Gly481	Water bridge
His480	π -cation interaction

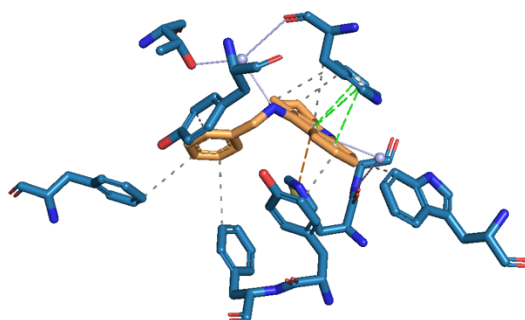


Figure 16. PLIP interaction visualization of the native co-crystallized ligand in the active-site region of AChE.

The PLIP visualization supported the residue-level interaction table by showing the spatial arrangement of the native ligand and its surrounding active-site residues.

Among these residues, Trp83 was particularly notable because it appeared in multiple interaction types, including hydrophobic interaction, water bridge, and π -stacking. This suggests that Trp83 may have an important role in stabilizing ligand binding within the AChE binding region. Other aromatic residues such as Tyr71, Phe330, Tyr370, Phe371, and Trp472 also contributed to the hydrophobic interaction environment of the native ligand.

The native ligand interaction profile showed that the AChE binding pocket contains several aromatic and hydrophobic residues that may support ligand stabilization. Therefore, these residues were used as reference residues when evaluating the docking results of limonene, L-carvone, carbaryl, and malathion. In the next step, the interaction profiles of the docked ligands were compared with the native ligand profile to determine whether they shared similar active-site residues.

4.6 Comparison of Amino Acid Interactions Between the Native and Docked Ligands

After analyzing the native co-crystallized ligand, the amino acid interaction profiles of the docked ligands were compared with the native ligand profile. This comparison was performed to evaluate whether limonene, L-carvone, carbaryl, and malathion interacted with residues located in the same or similar binding region of AChE. The main interaction types and interacting residues of each ligand are summarized in Table 8.

Table 8. Summary of protein–ligand interactions between AChE and the native and docked ligands.

Ligand	Main interaction types	Interacting residues
Native ligand (760-A-602)	Hydrophobic interaction, water bridge, π -stacking, π -cation interaction	Tyr71, Trp83, Phe330, Tyr370, Phe371, Trp472, Thr154, Gly481, His480
Limonene	Hydrophobic interaction	Tyr71, Glu80, Trp83, Tyr370, Tyr374, Trp472, Leu479
L-carvone	Hydrophobic interaction	Tyr71, Glu80, Trp83, Tyr370, Trp472
Carbaryl	Hydrophobic interaction, hydrogen bond, π -stacking	Tyr71, Glu80, Trp472, Tyr370, Trp83
Malathion	Hydrophobic interaction, hydrogen bond, salt bridge	Tyr71, Trp271, Leu328, Tyr370, Phe371, Phe440, Gly150, Ser238, His480

Table 8 shows that all docked ligands interacted with residues located in or near the native ligand-binding region of AChE. The native ligand formed hydrophobic interactions, water bridges, π -stacking, and π -cation interactions with several aromatic and active-site residues, including Tyr71, Trp83, Phe330, Tyr370, Phe371, Trp472, Thr154, Gly481, and His480. Limonene and L-carvone mainly showed hydrophobic interactions, suggesting that their binding was mostly supported by nonpolar contacts within the AChE binding pocket. Carbaryl showed a more diverse interaction profile, including hydrophobic interaction, hydrogen bond formation, and π -stacking. This diversity supported its stronger predicted binding affinity and lower K_i value. Malathion also showed multiple interaction types, including hydrophobic interaction, hydrogen bond, and salt bridge formation; however, its residue overlap with the native ligand was different from that of limonene, L-carvone, and

carbaryl. Therefore, Table 8 indicates that carbaryl had the strongest residue-level interaction profile, while the natural monoterpenes showed simpler but still relevant interaction patterns within the AChE binding region. To support the residue-level comparison, PyMOL docking pose images and Discovery Studio two-dimensional interaction diagrams of the docked ligands were included. PyMOL images show the spatial position of each ligand within the AChE binding region, while Discovery Studio diagrams support the interpretation of residue-level interaction patterns.

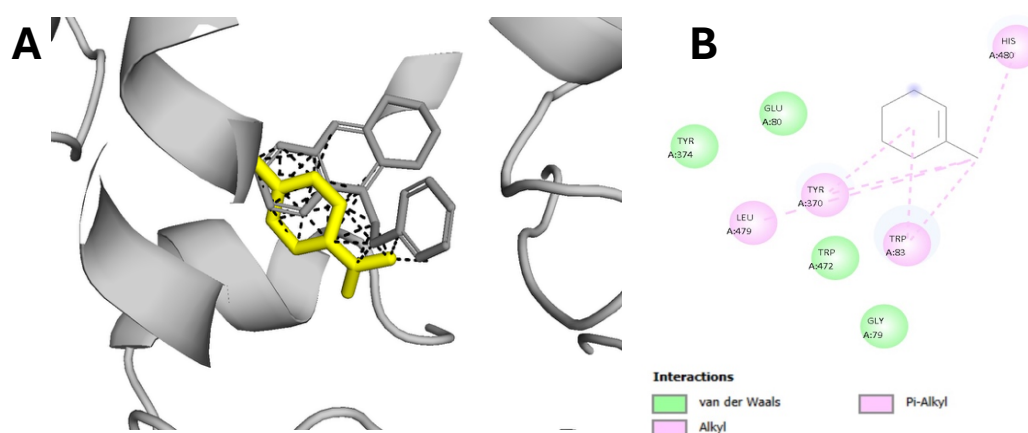


Figure 17. Docking pose and two-dimensional interaction diagram of limonene in the active-site region of AChE. A. The docking pose was visualized using PyMOL. B. 2D interaction diagram was generated using Discovery Studio Visualizer.

Limonene mainly showed hydrophobic interaction-related contacts in the AChE binding region. The Discovery Studio diagram indicated van der Waals, alkyl, and π -alkyl interactions with nearby residues, supporting its predicted placement within the binding pocket.

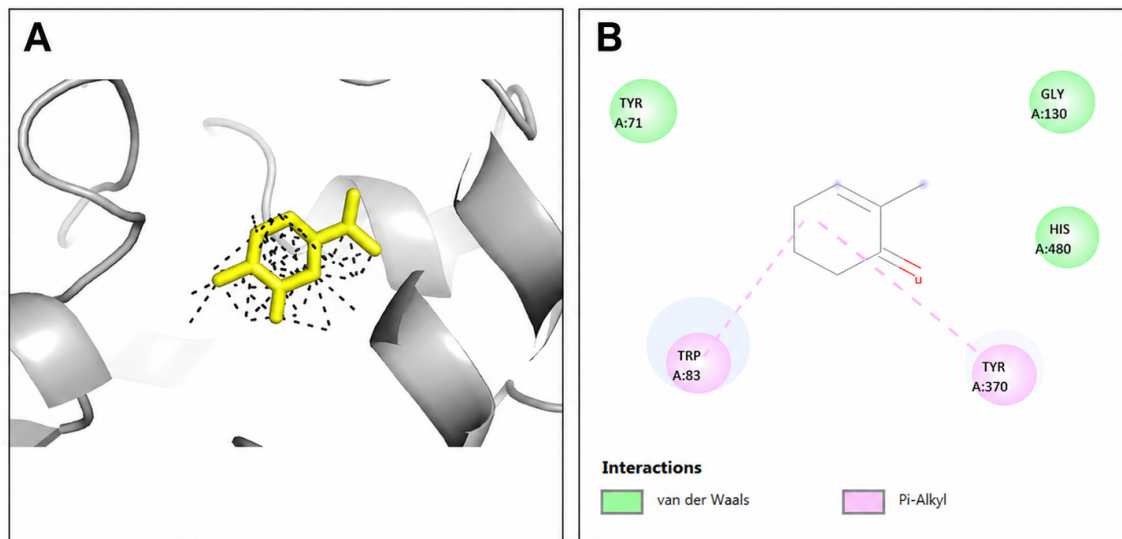


Figure 18. Docking pose and two-dimensional interaction diagram of L-carvone in the active-site region of AChE. A. The docking pose was visualized using PyMOL. B. The 2D interaction diagram was generated using Discovery Studio Visualizer.

L-carvone showed van der Waals and π -alkyl interaction-related contacts within the AChE binding region, supporting its predicted placement in the active-site environment.

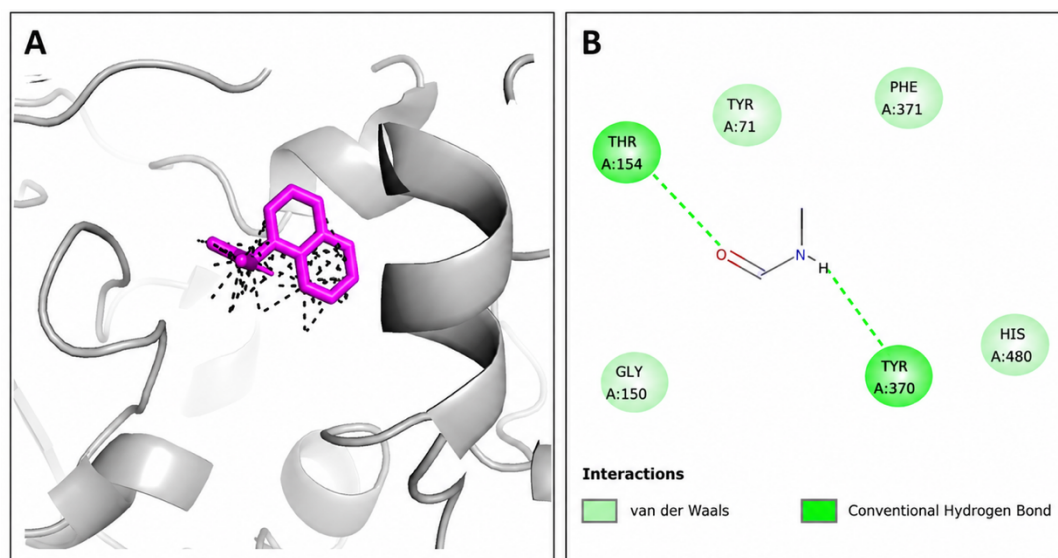


Figure 19. Docking pose and two-dimensional interaction diagram of carbaryl in the active-site region of AChE. A. The docking pose was visualized using PyMOL. B. The 2D interaction diagram was generated using Discovery Studio Visualizer.

Carbaryl showed hydrogen bond-related contacts and van der Waals interactions in the AChE binding region, supporting its stronger predicted binding profile compared with the other tested ligands.

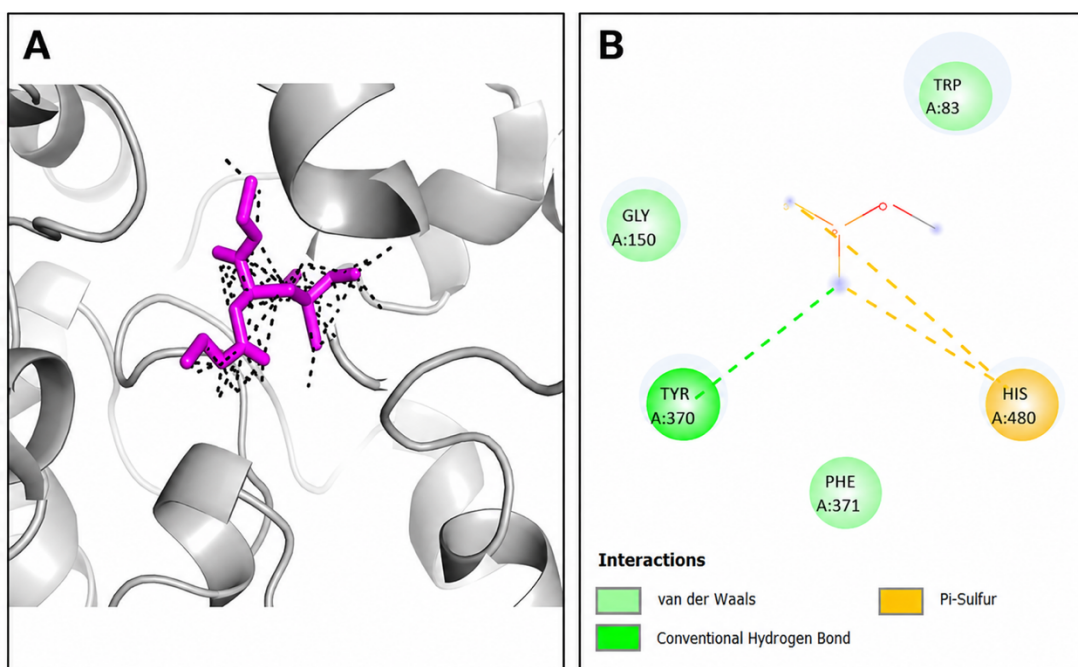


Figure 20. Docking pose and two-dimensional interaction diagram of malathion in the active-site region of AChE. A. The docking pose was visualized using PyMOL. B. The 2D interaction diagram was generated using Discovery Studio Visualizer.

Malathion showed interaction contacts within the AChE binding region, including van der Waals interactions, a conventional hydrogen bond, and a pi-sulfur interaction. This diagram supported the interpretation that malathion occupied a relevant region of the binding pocket, although its predicted binding profile was weaker than carbaryl.

The native ligand interacted with several aromatic and hydrophobic residues, including Tyr71, Trp83, Phe330, Tyr370, Phe371, and Trp472. These residues were used as reference points when evaluating the docked ligands. Among the natural compounds, limonene showed hydrophobic interactions with Tyr71, Glu80, Trp83, Tyr370, Tyr374, Trp472, and Leu479. Four of these residues, Tyr71, Trp83, Tyr370, and Trp472, were also observed in the native ligand interaction profile.

L-carvone also showed mainly hydrophobic interactions and interacted with Tyr71, Glu80, Trp83, Tyr370, and Trp472. Similar to limonene, it shared Tyr71, Trp83, Tyr370, and Trp472 with the native ligand. This overlap suggests that both natural monoterpenes were able to occupy a relevant part of the AChE binding region, although their interaction profiles were mainly hydrophobic.

Carbaryl showed a more diverse interaction profile than the natural ligands. In addition to hydrophobic interactions, carbaryl also formed a hydrogen bond and π -stacking interactions. It interacted with Tyr71, Glu80, Trp472, Tyr370, and Trp83, and shared Tyr71, Trp83, Tyr370, and Trp472 with the native ligand. This interaction overlap, together with the lowest binding energy and K_i value, supports the interpretation that carbaryl had the strongest predicted binding behavior among the tested ligands.

Malathion showed a mixed interaction profile that included hydrophobic interactions, hydrogen bonds, and a salt bridge. Its interacting residues were Tyr71, Trp271, Leu328, Tyr370, Phe371, Phe440, Gly150, Ser238, and His480. Among these residues, Tyr71, Tyr370, Phe371, and His480 were also found in the native ligand interaction profile. This indicates that malathion had partial overlap with the native binding environment, although its shared residue pattern was different from that of carbaryl and the natural monoterpenes.

The shared residues between the native ligand and the docked ligands are summarized in Table 9.

Table 9. Shared residues between the native ligand and docked ligands.

Ligand	Shared residues with native ligand
Limonene	Tyr71, Trp83, Tyr370, Trp472
L-carvone	Tyr71, Trp83, Tyr370, Trp472
Carbaryl	Tyr71, Trp83, Tyr370, Trp472
Malathion	Tyr71, Tyr370, Phe371, His480

Overall, the interaction analysis showed that all docked ligands shared at least some residues with the native ligand. Limonene, L-carvone, and carbaryl shared the same four residues with the native ligand: Tyr71, Trp83, Tyr370, and Trp472. However, carbaryl showed the strongest overall profile because its residue overlap was supported by the lowest binding energy, the lowest K_i value, and multiple interaction types. Malathion also interacted with

relevant native-ligand residues, but its interaction profile was less similar to the native ligand profile than carbaryl in terms of shared aromatic residues.

5. DISCUSSION

5.1 Comparison of Natural and Synthetic Compounds

The present study compared two natural monoterpenes, limonene and L-carvone, with two synthetic pesticide compounds, carbaryl and malathion, against acetylcholinesterase (AChE) using molecular docking and supportive *in silico* analyses. The main purpose of this comparison was not to claim that natural compounds are automatically safer or more effective than synthetic pesticides, but to evaluate whether selected natural compounds could show meaningful interaction patterns with the same enzyme target.

The docking results showed that carbaryl had the strongest predicted binding affinity among the tested ligands, with a binding energy of -8.489 kcal/mol. This result was expected because carbaryl is a carbamate pesticide and AChE is one of the major molecular targets of carbamate and organophosphate insecticides. In contrast, the natural compounds showed weaker but still relevant predicted binding values. L-carvone had a binding energy of -6.675 kcal/mol, while limonene showed -6.356 kcal/mol. These results suggest that the natural monoterpenes did not bind as strongly as carbaryl, but they were still able to interact with the AChE binding region.

Among the natural compounds, L-carvone showed a slightly stronger binding profile than limonene. This difference may be related to the presence of an oxygen-containing functional group in L-carvone, while limonene is a more hydrophobic hydrocarbon structure. The presence of a polar functional group may contribute to additional contact possibilities within the binding pocket. However, both natural compounds mainly showed hydrophobic interaction patterns, indicating that their binding behavior was largely supported by nonpolar contacts within the AChE active-site environment.

Malathion showed a binding energy of -6.611 kcal/mol, which was close to the value obtained for L-carvone. This finding is important because it suggests that L-carvone had a predicted binding affinity comparable to malathion under the docking conditions used in this study. However, the comparison should be interpreted carefully because binding energy alone does not fully determine biological activity, toxicity, or pesticide effectiveness. For this reason, docking results were evaluated together with K_i values, residue-level interactions, and ADMET-related predictions.

Overall, the results indicate that carbaryl remained the strongest predicted AChE binder, while L-carvone showed the most promising profile among the natural monoterpenes. Limonene showed the weakest binding among the tested ligands, but its interaction with several relevant residues suggests that it may still have potential for further screening. These findings are consistent with the general view that natural compounds may provide useful chemical structures for pesticide-related research, although their activity and safety must be evaluated scientifically rather than assumed from their natural origin alone (Isman, 2006; Regnault-Roger et al., 2012).

5.2 Interpretation of Binding Affinity, RMSD, and Ki Values

Binding affinity values were used as the first parameter for comparing the predicted interaction strength of the selected ligands with AChE. A more negative binding energy generally indicates a more favorable predicted ligand–receptor interaction. Based on this criterion, carbaryl showed the strongest binding profile, followed by L-carvone, malathion, and limonene. This ranking was also supported by the estimated K_i values, where carbaryl had the lowest K_i value and therefore the strongest predicted inhibitory potential.

The K_i results provided an additional way to interpret the docking scores. Carbaryl had an estimated K_i value of 0.60 μM , which was clearly lower than the values calculated for the other ligands. L-carvone and malathion had similar K_i values, 12.88 μM and 14.16 μM respectively, while limonene showed the highest K_i value with 21.79 μM . This pattern supported the binding energy results and suggested that carbaryl had the strongest predicted inhibitory potential, whereas L-carvone was the most favorable natural compound.

RMSD analysis was also useful for evaluating the consistency of alternative docking poses. Limonene and carbaryl showed relatively stable alternative poses, with mode 2 and mode 3 RMSD values mostly below or near 2 Å. This suggests that their predicted poses were relatively consistent around the best-ranked conformation. L-carvone showed one RMSD value slightly above 2 Å, while malathion showed a higher mode 2 RMSD value of 3.037 Å. This indicates that malathion had greater conformational variation in at least one alternative pose.

The higher variation observed for malathion may be related to its more flexible structure and higher number of rotatable bonds. Compared with limonene, L-carvone, and carbaryl, malathion has a larger and more flexible molecular structure, which may allow it to adopt different conformations inside the binding region. This flexibility can be useful for forming different types of contacts, but it may also reduce pose consistency. Therefore, the RMSD results support the idea that malathion should be interpreted more cautiously than carbaryl or limonene in terms of docking pose stability.

Although docking scores and K_i values provide useful comparative information, they should not be interpreted as direct experimental evidence of inhibition. Molecular docking is a predictive method and depends on receptor preparation, ligand conformation, grid selection, and scoring function limitations. Therefore, the binding affinity, K_i , and RMSD results in this study should be considered as computational screening outputs that require experimental validation in future studies (Trott and Olson, 2010; Eberhardt et al., 2021).

5.3 Evaluation of ADMET Prediction Results

The ADMET-related results provided an additional layer of comparison between the selected natural and synthetic compounds. These predictions were not used as experimental pharmacokinetic or toxicity data, but as supportive *in silico* screening results. The SwissADME results showed that all ligands had zero Lipinski rule violations, suggesting that their basic physicochemical properties were within acceptable ranges for small-molecule screening. However, important differences were observed in TPSA, rotatable bond number, gastrointestinal absorption, and blood–brain barrier permeability.

The BOILED-Egg model was particularly useful for visualizing these differences. Limonene, L-carvone, and carbaryl were positioned within the yellow region of the BOILED-Egg plot, indicating predicted blood–brain barrier permeability. In contrast, malathion was positioned on the right side of the plot and outside the main BOILED-Egg region. This

difference was important because it reflected the higher TPSA and higher number of rotatable bonds of malathion compared with the other ligands.

Malathion had the highest TPSA value, 138.26 Å², and the highest number of rotatable bonds, 11. These features suggest that malathion is more polar and structurally more flexible than the other tested compounds. Higher polarity may reduce passive membrane permeability, while increased molecular flexibility may reduce favorable passive absorption. Therefore, although malathion showed a docking affinity close to L-carvone, its ADMET-related profile should be interpreted as different rather than simply less favorable. From a drug-likeness perspective, low predicted gastrointestinal absorption and no predicted blood–brain barrier permeability may indicate limited passive absorption and brain penetration. However, in a pesticide-related context, these properties may also suggest a more limited passive systemic and central nervous system exposure profile in non-target organisms.

This finding is important because it shows why docking results should not be interpreted alone. If only binding energy were considered, malathion and L-carvone would appear relatively close. However, the BOILED-Egg model showed that their predicted pharmacokinetic profiles were different. L-carvone showed high predicted gastrointestinal absorption and BBB permeability, while malathion showed low predicted gastrointestinal absorption and no BBB permeability. This suggests that L-carvone and malathion showed clearly different passive absorption and permeability profiles, and these differences should be interpreted according to the biological purpose of the compound rather than as a simple “better or worse” ranking.

Limonene showed an interesting profile because it was predicted to be BBB permeant but had low predicted gastrointestinal absorption. This may be related to its very low TPSA and hydrophobic character. Carbaryl showed a strong docking profile and favorable predicted absorption/permeability characteristics, but as a synthetic carbamate pesticide, its toxicological relevance should still be interpreted in relation to its known AChE-inhibiting properties. Overall, the ADMET results supported the idea that computational evaluation should include both binding behavior and pharmacokinetic-related properties in order to provide a more balanced comparison.

5.4 Evaluation of Protein–Ligand Interaction Profiles

Protein–ligand interaction analysis was performed to understand whether the selected ligands interacted with residues located in or near the AChE binding region. The native co-crystallized ligand was first analyzed as a reference. The native ligand interacted with several residues, including Tyr71, Trp83, Phe330, Tyr370, Phe371, Trp472, Thr154, Gly481, and His480. These residues provided a reference profile for evaluating the docked ligands.

The interaction profiles showed that all docked ligands shared at least some residues with the native ligand. Limonene, L-carvone, and carbaryl shared Tyr71, Trp83, Tyr370, and Trp472 with the native ligand. This overlap suggests that these ligands occupied a relevant region of the AChE binding pocket. However, their interaction types differed. Limonene and L-carvone mainly showed hydrophobic interaction patterns, while carbaryl showed a more diverse interaction profile that included hydrophobic interactions, hydrogen bonding, and π -stacking.

The stronger predicted binding of carbaryl can be explained not only by its lower binding energy, but also by its interaction diversity and residue overlap with the native ligand. The presence of multiple interaction types may contribute to more stable binding within the

active-site region. This supports the interpretation that carbaryl had the strongest predicted binding behavior among the tested compounds. This result is also consistent with the known importance of AChE inhibition in the mechanism of carbamate pesticides (Costa, 2006; Gupta, 2011).

Limonene and L-carvone showed weaker binding than carbaryl, but their overlap with native-ligand residues is still meaningful. Their interaction with residues such as Tyr71, Trp83, Tyr370, and Trp472 suggests that they can occupy part of the AChE binding region. However, because their interactions were mainly hydrophobic, their predicted binding stability may be lower than that of carbaryl. Between the two natural compounds, L-carvone showed a slightly better binding energy and lower K_i value, supporting its more favorable predicted interaction profile.

Malathion showed partial overlap with the native ligand by sharing Tyr71, Tyr370, Phe371, and His480. Its interaction profile included hydrophobic interactions, hydrogen bonds, and a salt bridge according to PLIP analysis, while Discovery Studio visualization supported the presence of van der Waals contacts, hydrogen bonding, and pi-sulfur interaction. The interaction types identified by PLIP and Discovery Studio were not completely identical, but this is expected because different programs use different criteria for detecting protein–ligand contacts. Therefore, these tools were interpreted together rather than as conflicting outputs.

Overall, the interaction analysis supported the numerical docking results. Carbaryl had the strongest overall predicted profile due to its low binding energy, low K_i value, and multiple interaction types. L-carvone showed the best profile among the natural monoterpenes, while limonene showed weaker but still relevant binding-region contacts. Malathion interacted with relevant residues but showed a distinct ADMET-related profile and greater pose variation than carbaryl.

The Discovery Studio two-dimensional interaction diagrams provided an additional visual interpretation of the ligand–residue contacts. These diagrams were useful for showing interaction types such as van der Waals contacts, alkyl interactions, π -alkyl interactions, conventional hydrogen bonds, and pi-sulfur interactions in a more readable way. For limonene and L-carvone, the diagrams mainly supported hydrophobic interaction-related contacts, which was consistent with their mostly nonpolar monoterpene structures. Carbaryl showed hydrogen bond-related contacts together with van der Waals interactions, supporting its stronger predicted binding profile. Malathion showed van der Waals contacts, a conventional hydrogen bond, and a pi-sulfur interaction in the Discovery Studio diagram. Therefore, the Discovery Studio visualizations strengthened the residue-level interpretation by providing graphical support for the interaction patterns obtained from docking and PLIP analysis.

5.5 Evaluation of Natural Compounds as Alternative Pesticide Candidates

The results of this study suggest that natural monoterpenes may have potential as pesticide-related candidate molecules, but their role should be evaluated carefully. Limonene and L-carvone did not show stronger binding than carbaryl, but they interacted with residues located in the AChE binding region and shared several residues with the native ligand. This indicates that these compounds may have biologically relevant interaction potential, especially as preliminary candidates for further investigation.

Among the natural compounds, L-carvone appeared more promising than limonene. It showed a stronger binding energy, a lower K_i value, and a favorable predicted ADMET profile. Its binding affinity was also close to that of malathion, suggesting that L-carvone may

be a useful natural compound for further screening against AChE. However, this does not mean that L-carvone can directly replace synthetic pesticides. Field effectiveness, selectivity, formulation stability, environmental persistence, and experimental toxicity must be evaluated before any practical conclusion can be made.

Limonene showed weaker binding than L-carvone, but it still interacted with residues in the AChE binding region. Since limonene is a common monoterpene found in essential oils, its potential pesticidal activity may not depend only on strong AChE inhibition. It may also act through repellent effects, membrane disruption, or other mechanisms reported for essential oil components. Therefore, limonene may still be relevant in pest management research even if its predicted AChE binding profile is weaker than carbaryl or L-carvone (Nerio et al., 2010; Regnault-Roger et al., 2012).

The comparison with synthetic pesticides also shows that “natural” and “synthetic” should not be treated as simple categories of safe and harmful compounds. Carbaryl showed the strongest predicted binding profile, but strong AChE inhibition is also related to toxicological concern in non-target organisms. Malathion showed moderate docking performance but a clearly distinct predicted ADMET-related profile in this study. On the other hand, the natural compounds showed weaker AChE binding but more balanced predicted profiles in some parameters. This supports the idea that safer pesticide development requires a combined evaluation of activity, selectivity, pharmacokinetic-related properties, and toxicity risk.

The main limitation of this study is that it is based on computational predictions. Docking and ADMET analyses provide useful preliminary information, but they cannot replace experimental enzyme inhibition assays, toxicity tests, or biological activity studies. Therefore, future studies should include *in vitro* AChE inhibition assays, dose-response analysis, and toxicity evaluation on non-target organisms. In addition, molecular dynamics simulations could be used to examine the stability of ligand–AChE complexes over time.

In conclusion, the findings indicate that carbaryl showed the strongest predicted AChE binding, while L-carvone showed the most favorable result among the natural monoterpenes. Limonene also showed relevant active-site contacts, although its binding profile was weaker. Malathion had a binding affinity close to L-carvone, but its ADMET-related profile was clearly different due to high TPSA, high flexibility, low predicted gastrointestinal absorption, and lack of predicted BBB permeability. These results suggest that L-carvone and limonene may be considered preliminary natural candidates for further pesticide-related research, but experimental validation is necessary before drawing final conclusions about their inhibitory activity or practical use.

6. CONCLUSION

In this study, the binding behavior of two natural monoterpenes, limonene and L-carvone, and two synthetic pesticide compounds, carbaryl and malathion, was evaluated against acetylcholinesterase (AChE) using molecular docking and supportive *in silico* analyses. The results were interpreted by considering binding energy values, RMSD values, estimated K_i values, ADMET-related predictions, and protein–ligand interaction profiles.

Among all tested compounds, carbaryl showed the strongest predicted binding affinity toward AChE, with the lowest binding energy and the lowest estimated K_i value. This result was supported by its interaction profile, which included residue overlap with the native ligand and more diverse interaction types. Therefore, carbaryl appeared as the strongest predicted AChE binder in this computational comparison.

Among the natural compounds, L-carvone showed a more favorable profile than limonene. L-carvone had a stronger binding energy, a lower estimated K_i value, and a more favorable predicted ADMET profile compared with limonene. Its binding affinity was also close to that of malathion, suggesting that L-carvone may have meaningful interaction potential with the AChE binding region. Limonene showed the weakest binding affinity among the selected ligands, but it still interacted with residues located in or near the AChE active-site environment.

The ADMET prediction results showed that docking affinity alone is not sufficient for evaluating the selected compounds. This was especially clear for malathion. Although malathion showed a binding energy close to L-carvone, its BOILED-Egg position and SwissADME results indicated low predicted gastrointestinal absorption, no predicted blood–brain barrier permeability, high TPSA, and a high number of rotatable bonds. These results suggested that malathion had a clearly different ADMET-related profile compared with the other ligands. From a drug-likeness perspective, low predicted gastrointestinal absorption and no predicted blood–brain barrier permeability may indicate limited passive absorption and brain penetration. However, in a pesticide-related context, these properties may also suggest more limited passive systemic and central nervous system exposure in non-target organisms. Therefore, malathion’s ADMET-related profile should be interpreted according to the biological purpose of the compound rather than as a simple favorable or unfavorable result.

Protein–ligand interaction analysis also supported the docking results. Limonene, L-carvone, and carbaryl shared several residues with the native ligand, especially Tyr71, Trp83, Tyr370, and Trp472. Malathion also shared some native-ligand residues, including Tyr71, Tyr370, Phe371, and His480, but its interaction pattern was less similar to carbaryl and the natural monoterpenes. The use of PyMOL, PLIP, and Discovery Studio Visualizer helped support the interpretation of both docking poses and residue-level interaction patterns.

Overall, the findings suggest that carbaryl had the strongest predicted AChE binding profile, while L-carvone showed the most promising result among the natural monoterpenes. However, since this study was based on computational predictions, the results should be considered preliminary computational screening data. Further experimental studies, including in vitro AChE inhibition assays, toxicity evaluation, and biological activity tests, are required to confirm the inhibitory potential and practical pesticide-related relevance of limonene and L-carvone.

7. REFERENCES

1. Alavanja, M. C. R. (2009). Introduction: Pesticides use and exposure extensive worldwide. *Reviews on Environmental Health*, 24(4), 303–309.
2. Alavanja, M. C. R., & Bonner, M. R. (2012). Occupational pesticide exposures and cancer risk: A review. *Journal of Toxicology and Environmental Health, Part B*, 15(4), 238–263. <https://doi.org/10.1080/10937404.2012.632358>
3. Alengebawy, A., Abdelkhalek, S. T., Qureshi, S. R., & Wang, M. Q. (2021). Heavy metals and pesticides toxicity in agricultural soil and plants: Ecological risks and human health implications. *Toxics*, 9(3), 42. <https://doi.org/10.3390/toxics9030042>
4. BIOVIA, Dassault Systèmes. (2024). *BIOVIA Discovery Studio Visualizer*. Dassault Systèmes.
5. Colović, M. B., Krstić, D. Z., Lazarević-Pašti, T. D., Bondžić, A. M., & Vasić, V. M. (2013). Acetylcholinesterase inhibitors: Pharmacology and toxicology. *Current Neuropharmacology*, 11(3), 315–335. <https://doi.org/10.2174/1570159X11311030006>

6. Costa, L. G. (2006). Current issues in organophosphate toxicology. *Clinica Chimica Acta*, 366(1–2), 1–13. <https://doi.org/10.1016/j.cca.2005.10.008>
7. Daina, A., Michielin, O., & Zoete, V. (2017). SwissADME: A free web tool to evaluate pharmacokinetics, drug-likeness and medicinal chemistry friendliness of small molecules. *Scientific Reports*, 7, 42717. <https://doi.org/10.1038/srep42717>
8. Damalas, C. A., & Eleftherohorinos, I. G. (2011). Pesticide exposure, safety issues, and risk assessment indicators. *International Journal of Environmental Research and Public Health*, 8(5), 1402–1419. <https://doi.org/10.3390/ijerph8051402>
9. dos Santos, T. C., Gomes, T. M., Pinto, B. A. S., Camara, A. L., & Paes, A. M. A. (2018). Naturally occurring acetylcholinesterase inhibitors and their potential use for Alzheimer's disease therapy. *Frontiers in Pharmacology*, 9, 1192. <https://doi.org/10.3389/fphar.2018.01192>
10. Eberhardt, J., Santos-Martins, D., Tillack, A. F., & Forli, S. (2021). AutoDock Vina 1.2.0: New docking methods, expanded force field, and Python bindings. *Journal of Chemical Information and Modeling*, 61(8), 3891–3898. <https://doi.org/10.1021/acs.jcim.1c00203>
11. Freire, C., & Koifman, S. (2012). Pesticide exposure and Parkinson's disease: Epidemiological evidence of association. *NeuroToxicology*, 33(5), 947–971. <https://doi.org/10.1016/j.neuro.2012.05.011>
12. Gupta, R. C., Malik, J. K., & Milatovic, D. (2011). Organophosphate and carbamate pesticides. In R. C. Gupta (Ed.), *Reproductive and Developmental Toxicology* (pp. 471–486). Academic Press.
13. Hayden, K. M., Norton, M. C., Darcey, D., Østbye, T., Zandi, P. P., Breitner, J. C. S., & Welsh-Bohmer, K. A. (2010). Occupational exposure to pesticides increases the risk of incident AD: The Cache County Study. *Neurology*, 74(19), 1524–1530. <https://doi.org/10.1212/WNL.0b013e3181dd4423>
14. Isman, M. B. (2006). Botanical insecticides, deterrents, and repellents in modern agriculture and an increasingly regulated world. *Annual Review of Entomology*, 51, 45–66. <https://doi.org/10.1146/annurev.ento.51.110104.151146>
15. Jang, C., Park, J. W., Kim, S. K., Lee, S., & Kim, S. (2018). Identification of novel acetylcholinesterase inhibitors designed by pharmacophore-based 3D QSAR. *Scientific Reports*, 8, 14921. <https://doi.org/10.1038/s41598-018-33354-6>
16. Jokanović, M. (2018). Neurotoxic effects of organophosphorus pesticides and possible association with neurodegenerative diseases in man: A review. *Toxicology*, 410, 125–131. <https://doi.org/10.1016/j.tox.2018.09.009>
17. Kim, K. H., Kabir, E., & Jahan, S. A. (2017). Exposure to pesticides and the associated human health effects. *Science of the Total Environment*, 575, 525–535. <https://doi.org/10.1016/j.scitotenv.2016.09.009>
18. Kim, S., Thiessen, P. A., Bolton, E. E., Chen, J., Fu, G., Gindulyte, A., Han, L., He, J., He, S., Shoemaker, B. A., Wang, J., Yu, B., Zhang, J., & Bryant, S. H. (2016). PubChem Substance and Compound databases. *Nucleic Acids Research*, 44(D1), D1202–D1213. <https://doi.org/10.1093/nar/gkv951>
19. Landrum, G. (2024). *RDKit: Open-source cheminformatics software*. RDKit. <https://www.rdkit.org/>
20. Lionta, E., Spyrou, G., Vassilatis, D. K., & Cournia, Z. (2014). Structure-based virtual screening for drug discovery: Principles, applications and recent advances. *Current Topics in Medicinal Chemistry*, 14(16), 1923–1938. <https://doi.org/10.2174/1568026614666140929124445>

21. Meng, X. Y., Zhang, H. X., Mezei, M., & Cui, M. (2011). Molecular docking: A powerful approach for structure-based drug discovery. *Current Computer-Aided Drug Design*, 7(2), 146–157. <https://doi.org/10.2174/157340911795677602>
22. Mnif, W., Hassine, A. I. H., Bouaziz, A., Bartegi, A., Thomas, O., & Roig, B. (2011). Effect of endocrine disruptor pesticides: A review. *International Journal of Environmental Research and Public Health*, 8(6), 2265–2303. <https://doi.org/10.3390/ijerph8062265>
23. Mostafalou, S., & Abdollahi, M. (2013). Pesticides and human chronic diseases: Evidences, mechanisms, and perspectives. *Toxicology and Applied Pharmacology*, 268(2), 157–177. <https://doi.org/10.1016/j.taap.2013.01.025>
24. Mostafalou, S., & Abdollahi, M. (2018). The link of organophosphorus pesticides with neurodegenerative and neurodevelopmental diseases based on evidence and mechanisms. *Toxicology*, 409, 44–52. <https://doi.org/10.1016/j.tox.2018.07.014>
25. Mutunga, J. M. (2011). *Molecular, target-site, and toxicological studies of acetylcholinesterase insensitivity in the mosquito Anopheles gambiae* [Doctoral dissertation, Virginia Polytechnic Institute and State University]. Virginia Tech.
26. Nerio, L. S., Olivero-Verbel, J., & Stashenko, E. (2010). Repellent activity of essential oils: A review. *Bioresource Technology*, 101(1), 372–378. <https://doi.org/10.1016/j.biortech.2009.07.048>
27. Pang, Y. P. (2006). Novel acetylcholinesterase target site for malaria mosquito control. *PLoS ONE*, 1(1), e58. <https://doi.org/10.1371/journal.pone.0000058>
28. Protein Data Bank. (n.d.). *Acetylcholinesterase structure*, PDB ID: 6XYX. RCSB Protein Data Bank. <https://www.rcsb.org/structure/6XYX>
29. Regnault-Roger, C., Vincent, C., & Arnason, J. T. (2012). Essential oils in insect control: Low-risk products in a high-stakes world. *Annual Review of Entomology*, 57, 405–424. <https://doi.org/10.1146/annurev-ento-120710-100554>
30. Ribeiro-Filho, J., Leite, F. C., Costa, H. F., Calheiros, A. S., Torres, R. C., de Azevedo, C. T., de Carvalho, K. I. M., Figueiredo, J. G., Silva, R. O., & others. (2020). Carvone enantiomers differentially modulate IgE-mediated mast cell degranulation. *International Journal of Molecular Sciences*, 21(23), 9209. <https://doi.org/10.3390/ijms21239209>
31. Salentin, S., Schreiber, S., Haupt, V. J., Adasme, M. F., & Schroeder, M. (2015). PLIP: Fully automated protein–ligand interaction profiler. *Nucleic Acids Research*, 43(W1), W443–W447. <https://doi.org/10.1093/nar/gkv315>
32. Schrödinger, LLC. (2024). *The PyMOL Molecular Graphics System* (Version 3.1.6.1). Schrödinger, LLC.
33. Stanciu, G. D., Luca, A., Rusu, R. N., Bild, V., Chiriac, S. I. B., Solcan, C., Bild, W., & Ababei, D. C. (2020). Alzheimer’s disease pharmacotherapy in relation to cholinergic system involvement. *Biomolecules*, 10(1), 40. <https://doi.org/10.3390/biom10010040>
34. Sussman, J. L., Harel, M., Frolow, F., Oefner, C., Goldman, A., Toker, L., & Silman, I. (1991). Atomic structure of acetylcholinesterase from *Torpedo californica*: A prototypic acetylcholine-binding protein. *Science*, 253(5022), 872–879. <https://doi.org/10.1126/science.1678899>
35. Tanner, C. M., Kamel, F., Ross, G. W., Hoppin, J. A., Goldman, S. M., Korell, M., Marras, C., Bhudhikanok, G. S., Kasten, M., Chade, A. R., Comyns, K., Richards, M. B., Meng, C., Priestley, B., Fernandez, H. H., Cambi, F., Umbach, D. M., Blair, A., Sandler, D. P., & Langston, J. W. (2011). Rotenone, paraquat, and Parkinson’s disease. *Environmental Health Perspectives*, 119(6), 866–872. <https://doi.org/10.1289/ehp.1002839>

36. Torres, P. H. M., Sodero, A. C. R., Jofily, P., & Silva-Jr, F. P. (2019). Key topics in molecular docking for drug design. *International Journal of Molecular Sciences*, 20(18), 4574. <https://doi.org/10.3390/ijms20184574>
37. Trott, O., & Olson, A. J. (2010). AutoDock Vina: Improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of Computational Chemistry*, 31(2), 455–461. <https://doi.org/10.1002/jcc.21334>

CURRICULUM VITAE

Name and Surname: Miray CEYLAN

Student Number:

University: Halic University

Department: Molecular Biology and Genetics

Program: Undergraduate Program

Thesis Title:

Natural Terpenes vs Synthetic Pesticides: A Comparative Docking Study on
Acetylcholinesterase: Comparison of Limonene and L-Carvone with Carbaryl and Malathion

Supervisor:

Prof. Dr. Lecturer Mehmet Emin Uras